

UNIVERSITY OF MIAMI

PARAMETER OPTIMIZATION FOR THE SEPARATION AND EXTRACTION OF
DISTORTION PRODUCT OTOACOUSTIC EMISSIONS USING THE
SYNCHRONIZED SWEPT SINE METHOD

By

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A MASTERS RESEARCH PROJECT

Submitted to the Faculty of the University of Miami

in partial fulfillment of the requirements for

the degree of Master of Science

Coral Gables, Florida

May 2026

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Parameter Optimization for the Separation and
Extraction of Distortion Product Otoacoustic
Emissions Using the Synchronized Swept Sine Method

(May 2026)

Abstract of a thesis at the University of Miami.

Thesis supervised by Professor Christopher Bennett.

Distortion Product Otoacoustic Emissions (DPOAEs) are a vital clinical tool for assessing health of the cochlea and the Outer Hair Cells (OHCs). Traditional clinical methods rely on discrete tone pairs, which can be time-consuming and lack high-frequency resolution. This thesis expands on a novel approach - the Synchronized Swept Sine (SSS) method - to separate and extract DPOAE components. Using the SSS technique, the stimulus is deconvolved to produce a Virtual Impulse Response (VIR), allowing for the temporal separation of distortion products and their components. The objective of this work is to identify optimal parameters for the SSS method. Simulations were conducted across a range of sweep rates, alphas, and stimulus levels using the Verhulst 2012 cochlear model. Performance was evaluated in both noisy and non-noisy scenarios. Statistical analysis using a 2-way Analysis of Variance (ANOVA) was implemented to quantify the impact of parameters on component separation and Signal-to-Noise Ratio (SNR). The results show there is a trade-off between component separation and SNR. Although no single optimal parameter value was identified, the middle ground is the best way to maximize both. More simulations and live testing must be done to further validate these findings.

...

For my Parents, who gave me this opportunity.

For my sister, who inspires me.

For my brother, for his advice and empathy.

Acknowledgments

I want to express my gratitude to those who helped me achieve academic success. First and foremost, I would like to thank my family. Without them, I would not have been able to achieve all that I have. It was their compassion and support that gave me strength. Secondly, I would like to thank my committee: Dr. Christopher, Dr. Tom Collins, and Dr. Ozcan Ozdamar for their valuable insight and guidance throughout the process. However, I particularly would like to thank Dr. Christopher Bennett. His patience, expertise, and positive reinforcement helped see this through. Lastly, I would like to thank my peers. I will forever remember and cherish the time we shared.

Daniel Martens

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May 2026

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List of Abbreviations

- ANOVA: Analysis of Variance
- AWGN: Additive White Gaussian Noise
- BM: Basilar Membrane
- CRT: Critical Reflection Theory
- CWT: Continuous Wavelet Transform
- DP: Distortion Product
- DPOAE: Distortion Product Otoacoustic Emission
- ESS: Exponential Swept Sine
- HHFR: Higher Harmonic Frequency Response
- HHIR: Higher Harmonic Impulse Response
- IFFT: Inverse Fast Fourier Transform
- ImFR: Intermodulation Frequency Response
- ImIR: Intermodulation Impulse Response
- IMP: Intermodulation Product
- IR: Impulse Response

- LL: Long Latency
- LSF: Least Squares Fitting
- ms: Milliseconds
- OAE: Otoacoustic Emissions
- OD: Onset Decomposition
- OHC: Outer Hair Cell
- rms: root-mean-square
- SL: Short Latency
- SNR: Signal-to-Noise Ratio
- SSS: Synchronized Swept Sine
- VIR: Virtual Impulse Response

Chapter 1

Introduction to DPOAEs

Distortion product otoacoustic emissions (DPOAEs) are acoustic signals generated within the cochlea which can be measured in the ear canal with a probe. These acoustic signals are created by an interaction between two stimulus tones, f_1 and f_2 such that $f_2 > f_1$. The frequency of the distortion product can be defined as:

$$f_{DP} = mf_1 + nf_2$$

where m and n are integers. In theory, there are infinitely many possible distortion products (DPs), but the most important is the cubic distortion product, $2f_1 - f_2$ [1]. This is due to the strong mechanical interaction in the region where the traveling waves evoked by f_1 and f_2 overlap, a phenomenon consistent with cochlear macromechanical modeling [2]. Other DPs that will be analyzed in this optimization study are $2f_2 - f_1$ and $3f_1 - 2f_2$. DPOAEs are considered a by-product of the electromechanical force of the outer hair cell (OHC) that forms the basis of cochlear amplification. Thus, by measuring the amplitude of the DP, we can assess the health status of the cochlea and the OHCs. A reduced or absent amplitude would imply cochlear damage, which can result in hearing loss or, in more serious cases, deafness [3].

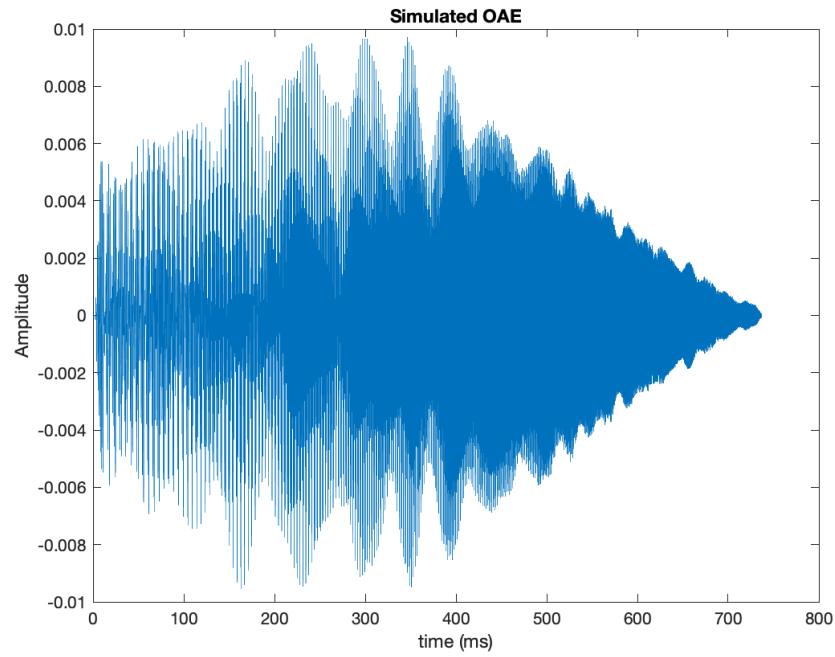


Figure 1.1: An image of a DPOAE generated using a cochlear simulator in MATLAB.

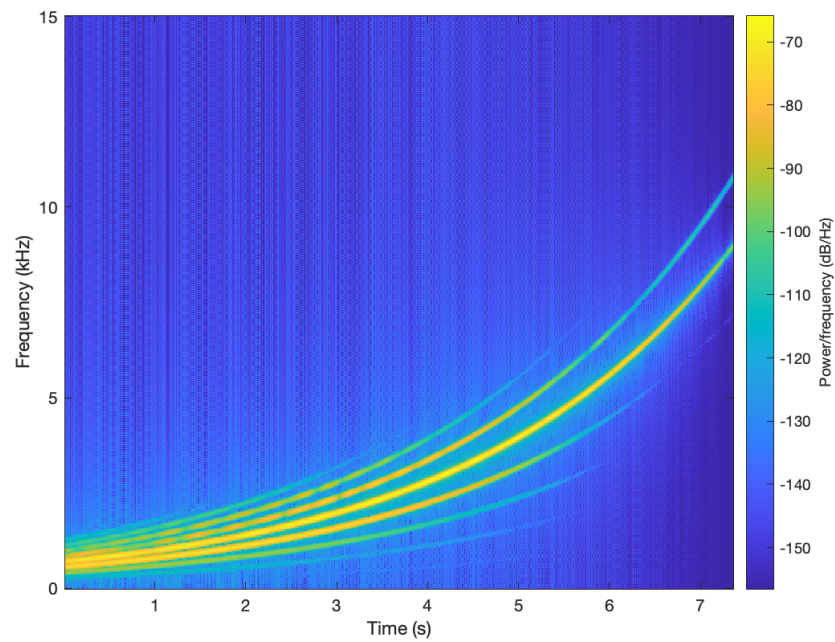


Figure 1.2: The spectrogram of a DPOAE to visualize the frequency content of a DPOAE.

1.1 Origin of Distortion Product

The energy of the DP originates in the region of maximum overlap of the traveling waves excited by the two primary tones. This primary generation site is generally located near the tonotopic place of the higher-frequency tone, f_2 [4]. It is the stereociliary transduction of the OHCs that produces the mechanical distortion that acts as the primary source for DPOAEs [5].

The energy generated from distortion travels in two directions: basally (backward) toward the ear canal and apically (forward) toward the cochlear apex [6]. The forward-traveling wave moves toward the tonotopic place of the DP frequency, and once it reaches that location, it is coherently reflected back toward the ear canal and mixes with the backward-traveling wave. However, for measurement purposes, the backward travel component is emphasized since it contains the distortion energy rather than the reflected energy. Thus, separation of the mixed components is necessary to analyze the backward-traveling wave.

1.2 Components of Distortion Products

However, we can further dissect the generation of DPOAEs by analyzing their individual components. The DPOAE measured in the ear canal is not derived from a single source, but rather represents the vector sum of at least two major components: a distortion component and a reflection component. These components are generated in distinct cochlear regions via fundamentally different mechanisms.

The interference between these two components, which alternates between constructive and destructive summation, gives rise to the characteristic quasi-periodic variations in DPOAE amplitude and phase known as DPOAE fine structure [7]. This spectral microstructure manifests itself as a series of peaks and notches across the DPOAE spectrum, resembling a comb filter.

1.2.1 Distortion Component

The distortion component is considered to be the primary source, that is, the more dominant of the two. It originates directly from the nonlinear interaction of the two primary tones in the traveling-wave overlap region, primarily near the f_2 tonotopic place. Distortion energy generated in this region propagates basally toward the ear canal as a backward-traveling wave, which constitutes the dominant contribution of the distortion component measured in the ear canal.

Because this region shifts systematically with stimulus frequency, the distortion source moves accordingly and is therefore described as wave-fixed. When the DPOAE phase is plotted versus the DP frequency with fixed α , the ratio of f_2/f_1 , the distortion component exhibits a relatively flat phase gradient, which corresponds to minimal delay, approximately 1-2 ms [8]. This is why the distortion component is also referred to as the short-latency (SL) component.

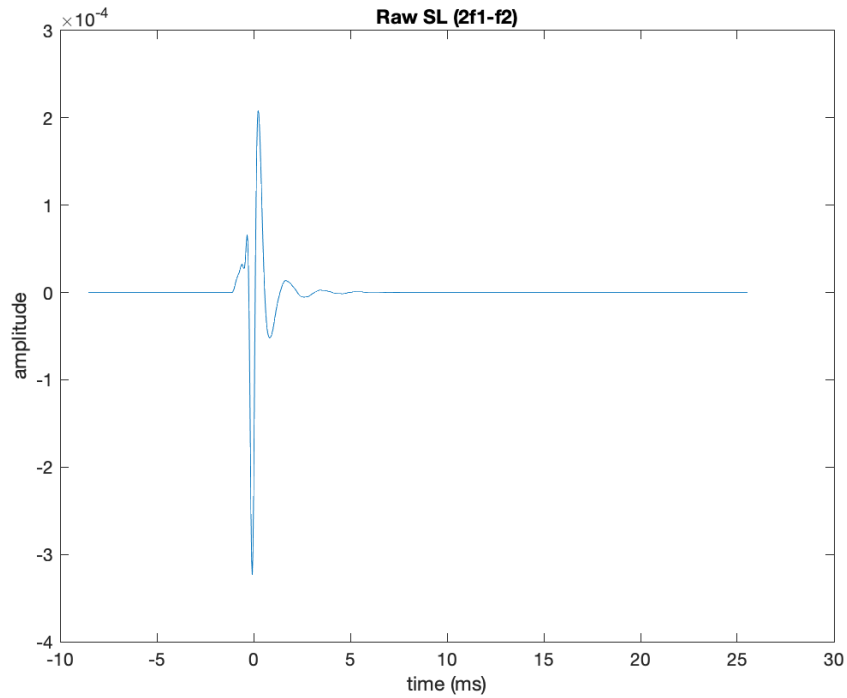


Figure 1.3: An image of the distortion component for the $2f_1 - f_2$ DP

1.2.2 Reflection Component

The reflection component is often referred to as the secondary source and is generated through a fundamentally different mechanism known as coherent reflection, which can be described by the Coherent Reflection Theory (CRT). Distortion energy is initially generated in the region of maximum overlap between the f_1 and f_2 traveling waves, which is located near the f_2 tonotopic place. For distortion products with frequencies lower than the f_2 tone, such as the cubic distortion product ($2f_1 - f_2$), the forward-traveling wave propagates apically and undergoes coherent scattering near the characteristic place of the DP.

At this location, the forward-traveling wave is scattered and reflected basally due to tiny, spatially dense, place-fixed impedance irregularities, which are then amplified by the OHCs [9]. The irregularities that cause this reflection are fixed along the basilar membrane (BM), leading this source to be described as place-fixed. This, in essence, is the Coherent Reflection Theory.

In further contrast to the distortion component, the reflection component is characterized by rapid phase variation, which implies a steep phase gradient. This steep phase gradient corresponds to a substantial delay, approximately 4–12 ms, which is why the reflection component is often referred to as the long-latency (LL) component.

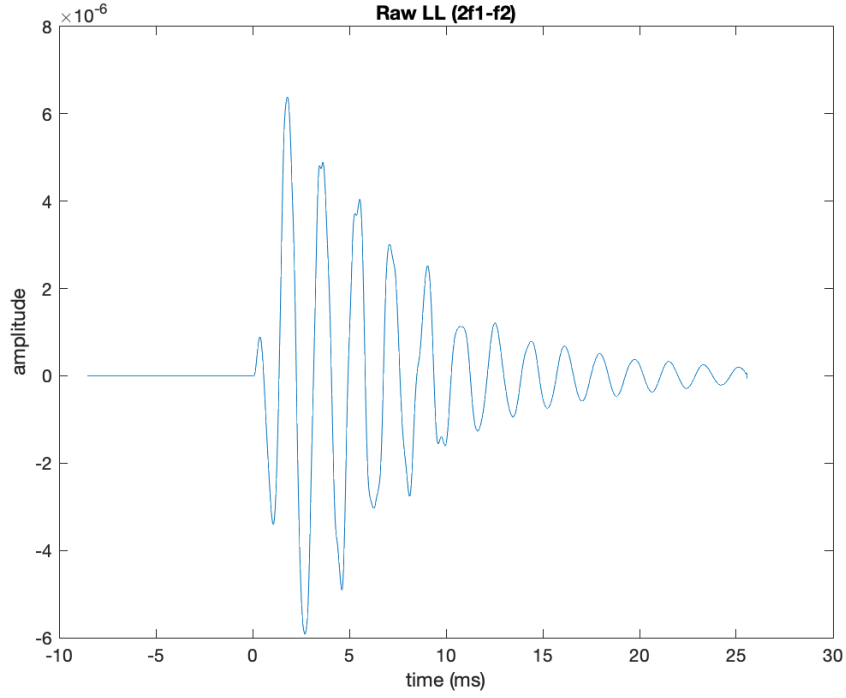


Figure 1.4: An image of the reflection component for the $2f_1 - f_2$ DP.

1.3 Component Characterization and Separation Challenges

When measuring the amplitude of the DPOAE, we are primarily interested in the amplitude of the SL component because it dominates the DPOAE level. Thus, the amplitude of these distortions provides a noninvasive metric of the ear's internal nonlinearity, which is a hallmark of a healthy cochlea [10]. A higher amplitude means that a stronger distortion is produced, implying a healthier cochlea. The strength of these distortions also signals that the OHCs are functioning properly, which is an additional indicator of good cochlear health.

However, there is a certain degree of overlap between the SL and LL components that can cause interference and complicate the reliable interpretation of measurements. This makes the isolation and extraction of the SL component of central interest for both clinical and

research applications. Although complete separation of the components is not possible, it can be optimized.

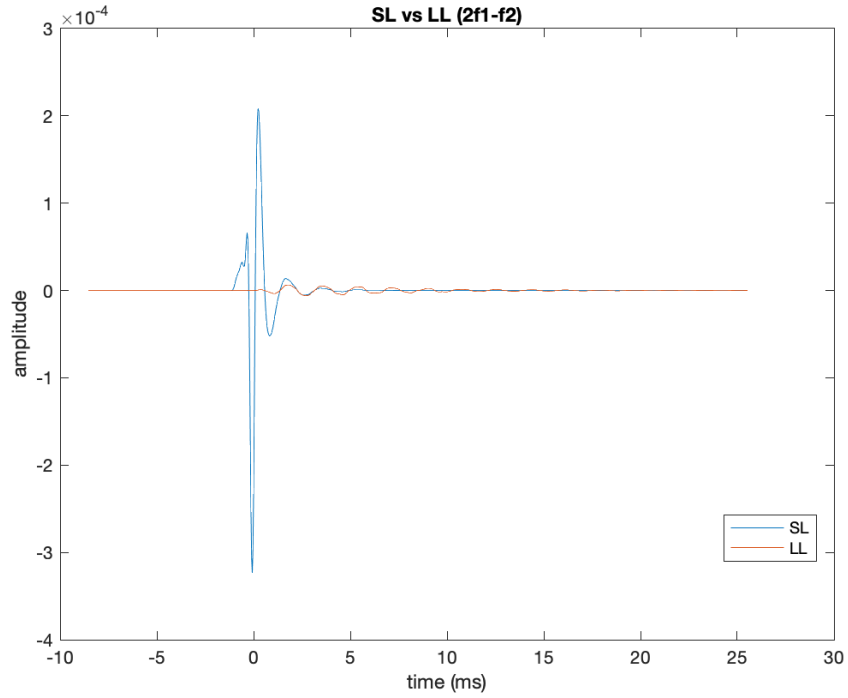


Figure 1.5: An image of the distortion and reflection components plotted together. This illustrates the temporal proximity of the two components which causes them to overlap.

Chapter 2

Literature Review

This section aims to survey the past and current state of research on DPOAE component separation and extraction. It discusses the development and theory of novel techniques, specifically the synchronized swept-sine method, which was formally developed by Novák et al. [11] for nonlinear system identification, revised in Novak et al. [12], and applied to DPOAEs in Vencovský et al. [13]. In addition, this section provides an overview and an approximate timeline of previously developed techniques, including the current state-of-the-art method, the least-squares fitting (LSF) method.

The literature review is broken down into three sections:

1. Foundations and Timeline of Prior Methods
2. Introduction to Synchronized Swept Sines
3. Analysis of SSS Parameters

2.1 Foundations and Timeline of Developed Methods for Component Separation

The history of component separation in DPOAEs is closely linked to the development of the two-source model of cochlear function, which explains the variability observed in DPOAE measurements. This model posits that the total DPOAE measured is the vector sum of two major components: the primary nonlinear distortion component and the secondary coherent reflection component. This two-source hypothesis, formalized by Shera and Guinan [14] and based on earlier concepts proposed by Kemp and Brown [15], explains the quasi-periodic DPOAE fine structure observed in DP-grams, a visual representation that plots DPOAE amplitude versus f_2 frequency, as a result of alternating constructive and destructive interference between the two components.

Because the distortion and reflection components exhibit strongly different phase-gradient behavior, they can be separated in the time domain using a variety of distinct methods.

2.1.1 Local vs. Global Approaches

These techniques can be broadly classified into two categories: local and global approaches. Local methods utilize temporal or pointwise analysis, meaning they achieve separation based on the time course of the response. Global methods utilize frequency-domain analysis because they require obtaining DPOAE data with high frequency resolution collected across a wide frequency range via a DP-gram. This distinction is useful for understanding both the diversity of implementations and the practical constraints that shaped the evolution of DPOAE separation techniques.

2.1.2 Classical Methods

Initial efforts to separate these components relied on two main classical methods: selective suppression, a local approach, and time-domain analysis, which incorporates both local and global approaches.

Selective Suppression (Local)

The selective suppression method uses a third suppressor tone, f_3 , presented near the DP frequency, f_{DP} , to suppress the reflection component, ideally leaving the distortion component intact [16]. The goal is to obtain an estimate of the distortion component uncontaminated by signal from the reflection component. While studies have shown this method can yield results similar to other techniques, its clinical utility is often limited.

The primary obstacle is finding an appropriate suppressor level that effectively suppresses the reflection source across a wide range of primary levels and in all subjects. Because the strength of the reflection component varies significantly between individuals, there is no universal suppressor level. Large-scale studies, such as [17], discovered that using fixed-level suppressors failed to improve the capacity of DPOAEs to accurately predict hearing loss compared to other methods. Searching for optimal suppressor parameters for each patient is time-consuming and therefore unattractive for rapid clinical use.

IFFT Analysis (Global)

Various time-domain methods have been proposed and used over the years. One such approach involves using the inverse fast Fourier transform (IFFT). This technique measures the DPOAE amplitude and phase across a frequency range with high resolution, that is, through a DP-gram. The IFFT transforms this frequency-domain data into the time domain, where the components naturally separate because of their different latencies. These separated parts can then be individually selected and windowed, a process known as smoothing or source

unmixing [18].

However, obtaining the required high-resolution data for IFFT analysis using discrete tones is extremely time-consuming. Due to the enormous measurement effort required, the IFFT approach is unsuitable for clinical applications.

Onset Decomposition (Local)

Another time-domain approach involved analyzing the onset of DPOAEs evoked by pulsed stimuli, often pulsing the f_2 tone while f_1 remained continuous. Experimental evidence demonstrated that the SL component reaches its steady state faster than the LL component builds up and begins to interfere. This temporal difference allows extraction of the primary source via methods such as onset decomposition (OD), which samples the DPOAE envelope at a precise, predefined time instant immediately following the onset.

In essence, the OD method can quickly separate the primary source from a single frequency point without measuring an entire DP-gram [19]. However, a significant drawback of the OD method is its underestimation of amplitude by approximately 2–2.3 dB, since it samples the DPOAE before it reaches steady state. Furthermore, the OD method uses a pointwise estimation, meaning the resulting estimate is highly sensitive to noise. Although considered an improvement over the IFFT approach, OD is still not considered ideal for clinical use.

Time-Frequency Domain Filtering (Global)

An additional classical approach used by Moleti et al. [20] is time-frequency domain filtering using the continuous wavelet transform (CWT). This technique decomposes OAE signals using frequency-scaled and time-shifted versions of the mother wavelet. By mapping OAE signals into a time-frequency representation, the different components occupy distinct regions in the time-frequency plane. Selective masking within this domain allows efficient isolation and reconstruction of individual components, improving signal-to-noise ratio (SNR) and

revealing detailed cochlear delay-frequency relationships.

This process proved highly effective for improving SNR and revealing detailed frequency-dependent cochlear delay relationships. While initially considered an improvement over other approaches, it has some limitations. One such limitation is that the window lengths are fixed to frequency location, which makes the method prone to artifacts [21]. Another limitation is redundancy, since the CWT produces an overcomplete representation, resulting in increased computational cost.

2.1.3 Transition to Modern Approaches

More recently, a major shift occurred with the introduction of swept-tone methods using continuously sweeping primaries. Talmadge and Long first presented this in 2004 before publishing it in 2008. The swept-tone technique allows for dramatically faster acquisition of high-resolution data necessary for source separation more efficiently than preceding discrete-tone methods. For example, a high-resolution DPOAE can be recorded in roughly 2.7 minutes, compared to 40 minutes for comparable discrete-tone resolution.

Swept-tone analysis is also more noise-robust than previous techniques. While traditional synchronous averaging cannot reduce coherent noise, swept-tone analysis uses an inverse sweep, which is more effective at reducing both random and phase-locked additive noise. Because the stimulus energy is distributed over a much longer duration than a click, the SNR is significantly improved. It has also shown to be more effective in component separation and extraction [22].

As such, the swept-tone paradigm is primarily considered a global approach, though certain implementations incorporate local analyses.

The least-squares fitting (LSF) procedure works by segmenting the recorded ear-canal waveform into short analysis windows. Then a mathematical model of the primary stimulus tones and the target DPOAE frequency is fit to these segments. The algorithm calculates the amplitude and phase of these components by minimizing the sum of squared residuals,

that is, the difference between the theoretical model and the actual measured pressure. A useful way to conceptualize the LSF is as a moving bandpass filter whose center frequency continuously tracks the instantaneous frequency.

An important insight from this approach was that component separation could be achieved directly through analysis parameters [23]. By using a rapid sweep rate, for example, 0.5 octaves/s, approximately 2 s/octave, along with a short analysis window, the LL component falls outside the time range of the LSF filter, effectively eliminating it from the measurement. This manipulation allows for rapid estimation of an uncontaminated distortion component.

The LSF variation of the swept-tone method opened the door for efficient measurement of distortion and reflection components in the same subject, a concept that was further developed in another variation of the swept-tone approach known as the synchronized swept-sine (SSS) method.

2.2 Introduction to Synchronized Swept Sines

The synchronized swept-sine (SSS) technique represents a refinement of swept-tone methodology, adapted from approaches originally developed to characterize general nonlinear acoustic systems. In this method, the stimulus consists of two exponentially swept primary tones, generated using specifically adjusted formulas that guarantee tight synchronization of the two continuously sweeping primaries. This synchronization ensures that specific DPs generated by nonlinearities occur at well-defined, time-locked frequency trajectories throughout the sweep.

The ear-canal response to these synchronized sweeps is deconvolved, yielding an impulse response (IR), which is a time-domain representation of frequency-dependent DPs. In the IR, individual DPs appear at different time locations instead of overlapping. This temporal separation arises because each nonlinear component contributes energy with a characteristic group delay that depends on both the order of the nonlinearity and the physical mechanism

generating the response. During deconvolution, these different delays are mapped onto different arrival times in the impulse response, effectively unfolding the nonlinear responses.

In cochlear measurements, this time-domain separation has a direct physiological interpretation. The SL and LL components of the DPOAE originate from different cochlear processes and therefore exhibit markedly different latencies: the SL component is associated with rapid, near-overlap interactions of the primary waves, whereas the reflection component involves longer cochlear travel and place-fixed reflection. As a result, these components appear at clearly separated time positions in the impulse response [12].

The SSS method thus provides a fast, high-resolution means of isolating specific DPOAE components with a continuous-sweep paradigm, combining the efficiency of swept-tone stimulation with the interpretability of time-resolved nonlinear system identification.

2.2.1 History of SSS Development

Engineers began using chirps, that is, swept sines, for precision acoustic measurements following World War II, with applications in radio engineering. Over many years, however, the swept-sine method was developed and refined. It gradually became more pertinent in other fields, moving from general acoustics and audio engineering to specialized cochlear measurement.

Pioneering work on swept-tone analysis was first theorized for audio systems by Craven and Gerzon [24] and [25]. Farina [26] and Farina [27] formally developed and applied the exponential swept-sine (ESS) technique. Farina’s work demonstrated the ability of the exponential sweep to simultaneously measure the linear impulse response and separate the impulse responses for each harmonic distortion via a process he called nonlinear convolution. Prior to its development, methods such as maximum length sequence (MLS) and time-delay spectrometry (TDS) were used to measure impulse responses of linear time-invariant (LTI) systems. However, these methods struggled when faced with nonlinearity or time variance, whereas Farina’s method succeeded.

Researchers soon identified the potential benefits of such an approach in analyzing cochlear function via otoacoustic emissions. Conventional discrete-tone OAE measurements were slow and cumbersome, making the speed of the swept-tone approach especially attractive [28]. Talmadge and Long were among the first to apply swept tones to measure DPOAEs, although they used LSF to analyze the results, since SSS had not yet been formally developed at the time. Choi et al. [29] introduced the use of continuously swept frequency or level stimuli combined with digital heterodyne analysis for measuring stimulus-frequency OAEs (SFOAEs) efficiently and with high resolution. Bennett et al. [30] formally developed the swept-tone technique for use in the human ear and were able to compare the results to transient-evoked OAEs (TEOAEs).

Novák et al. [11] reformulated the ESS theory, providing a specific mathematical formula for SSS generation. In 2015, Novak et al. [12] revised the method further, showing that if the signal is not correctly synchronized, the phases of the resulting higher harmonic impulse responses become dependent on the input signal, resulting in incorrect estimation. This refinement, ensuring phase accuracy, became known as the SSS theory. Vencovský et al. [13] subsequently adapted SSS for impulse extraction, specifically DPOAEs, from responses evoked by two simultaneous swept sines. It was shown to be comparable to the LSF method but with lower computational expense. Current work is being conducted to further optimize the SSS technique.

2.2.2 Distinguishing Two Types of SSS

Before discussing the theory behind SSS, it is important to note that it can be categorized into two groups: SSS with deconvolution and SSS without deconvolution. The distinction between these approaches primarily involves how the signal is processed to separate the linear response from nonlinear distortion and how background noise is handled.

Methods that do not use deconvolution, most notably LSF, analyze the recorded sweep directly as a function of time and frequency. For example, LSF models the emission as it

varies across time by fitting a short snippet of the known stimulus frequency to segments of the recording. It essentially functions as a moving bandpass filter. The components are then separated by manipulating the duration of the analysis window. A short window captures the entire DPOAE, including fine structure, whereas a long window isolates the SL component. Using prior knowledge of the signal, LSF is able to generate a noise-robust model that is less susceptible to artifacts.

When deconvolution is used, the SSS method follows a signal-processing strategy called nonlinear convolution. This method transforms a long-duration sweep into a virtual impulse response. The response is termed virtual because it applies the mathematical framework of linear system theory to a nonlinear system. Similarly to how convolution corresponds to multiplication in the frequency domain, deconvolution corresponds to division in the frequency domain. However, what most defines the SSS method using deconvolution is that it packs the response into a series of higher harmonic impulse responses or impulse responses that appear as distinct peaks at time-shifted intervals. The next section will examine the theory of the SSS method that uses deconvolution as it was used in [13].

2.2.3 Mathematics and Theory of the SSS

The SSS method is a sophisticated evolution of the ESS method, which was used for the fast and high-resolution measurement of nonlinear systems. It exhibits unique properties that allow for rapid analysis of the amplitude and phase of frequency-dependent DPs. While traditionally used for analyzing higher harmonics, it was later adapted in Vencovský et al. [13] to analyze IMPs. This section discusses the mathematics of the SSS and how it applies to distortion products.

Defining a SSS

A SSS signal is a special case of an ESS signal defined as:

$$s(t) = \sin(\phi(t)) \quad (2.1)$$

$$\phi(t) = 2\pi f_a L \exp\left(\frac{t}{L}\right) \quad (2.2)$$

$$L = \frac{T}{\ln(f_b/f_a)} \quad (2.3)$$

Here, ϕ corresponds to phase, and the coefficient L is the sweep-rate coefficient, which takes the following input parameters: start frequency f_a , stop frequency f_b , and a time vector t , in seconds, ranging from $t = 0$ to $t = T$.

A SSS signal has a unique property regarding higher harmonics, that is, positive integer multiples of a fundamental frequency. Multiplication of phase by a positive integer m corresponds to the formation of the m th harmonic of the swept sine, which is further equivalent to a time shift of the phase by Δt_m :

$$m\phi(t) = \phi(t - \Delta t_m) \quad (2.4)$$

$$\Delta t_m = -L \ln(m) \quad (2.5)$$

$$s_m(t) = \sin(m\phi(t)) = \sin(\phi(t - \Delta t_m)) \quad (2.6)$$

This harmonics property can be extended to intermodulation DPs so that the system's impulse response, $h(t)$, can be derived.

Extracting the IR from the Intermodulation Distortion

When simultaneously inputting pure sine waves, with respective frequencies f_1 and f_2 , into a nonlinear system, the input of the system, $x(t)$, is the summation of two synchronized swept sines, one for each respective frequency:

$$s(t) = s_1(t) + s_2(t) \quad (2.7)$$

$$s_1(t) = \sin(\phi_1(t)) \quad (2.8)$$

$$s_2(t) = \sin(\phi_2(t)) \quad (2.9)$$

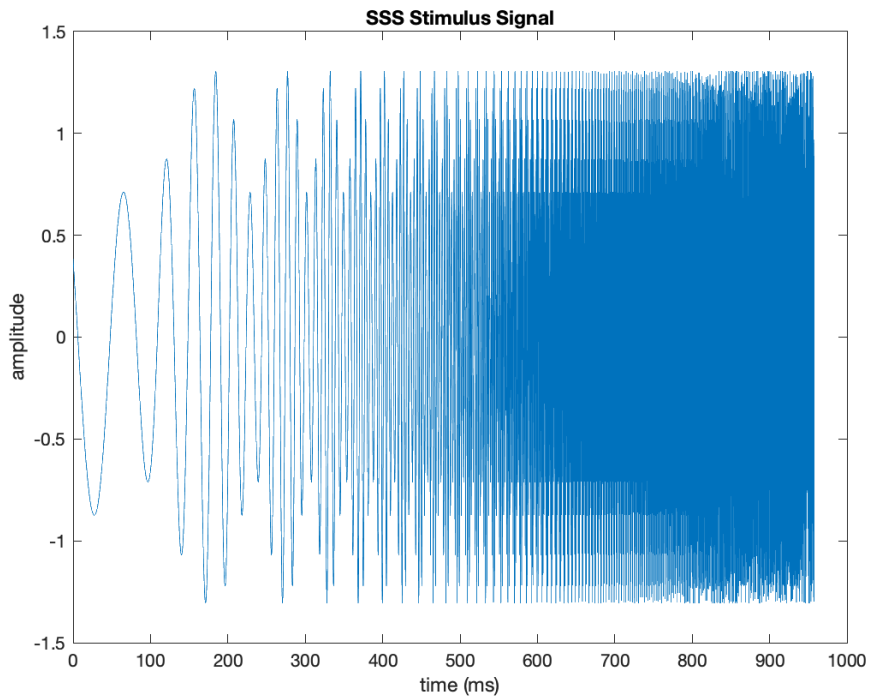


Figure 2.1: This figure illustrates what the SSS stimulus looks like in the time domain. The f_2 sweep is scaled in amplitude before being added to the f_1 sweep.

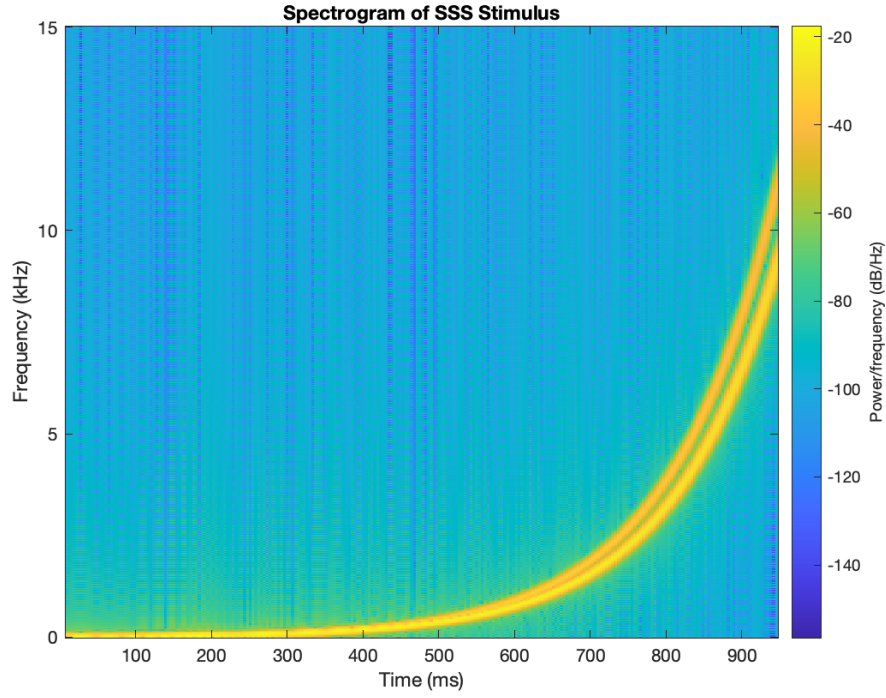


Figure 2.2: A spectrogram of the SSS stimulus from the figure above.

The output consists of both harmonics and intermodulation DPs. The derived frequencies of the DPs are restated below:

$$f_{m,n} = mf_1 + nf_2 \quad \text{where } m, n \in \mathbb{Z} \quad (2.10)$$

$f_{m,n}$ is a harmonic if $m = 0$ or $n = 0$.

By factoring out f_1 , we can write this equation in terms of f_1 and α , the ratio of the two input frequencies:

$$f_{m,n} = f_1 \left(m + \frac{f_2}{f_1} n \right) \quad (2.11)$$

$$f_{m,n} = f_1(m + \alpha n) \quad (2.12)$$

This indexing system is used not only for the frequency components but also for indexing

swept-sine signals and their ImIRs. Similarly to harmonics, intermodulation distortions are also frequency-dependent. Each distortion has its own amplitude $A_{m,n}$ and phase $\phi_{m,n}$, with m and n indicating the distortion product. The frequency response for each distortion is defined as

$$H_{m,n}(f) = A_{m,n}(f) \exp(\phi_{m,n}(f)) \quad (2.13)$$

By taking the inverse fast Fourier transform of the frequency response, we derive the impulse response of the individual distortion:

$$h_{m,n}(t) = \mathcal{F}^{-1}(H_{m,n}(f)) \quad (2.14)$$

Here, $\mathcal{F}$ denotes the Fourier transform, and $\mathcal{F}^{-1}$ denotes the inverse Fourier transform.

Vencovský et al. [13] shows that Equations (4), (5), and (6) can be applied to distortion products. The analogous form of those equations is as follows:

$$(m + \alpha n)\phi_1(t) = \phi_1(t - \Delta t_{m,n}) \quad (2.15)$$

$$\Delta t_{m,n} = -L \ln(m + \alpha n) \quad (2.16)$$

$$s_{m,n}(t) = \sin((m + \alpha n)\phi_1(t)) = \sin(\phi_1(t - \Delta t_{m,n})) \quad (2.17)$$

Thus, the output signal $y(t)$ can be described as the sum of all harmonic and intermodulation distortions. The final equation for the output can be expressed as a convolution of the first stimulus signal, $\sin(\phi_1(t))$, and an impulse response $h(t)$, which consists of ImIRs delayed by $\Delta t_{m,n}$:

$$y(t) = \sin(\phi_1(t)) * h(t) \quad (2.18)$$

$$h(t) = \sum_m \sum_n h_{m,n}(t - \Delta t_{m,n}) \quad (2.19)$$

Once the output signal is obtained, $h(t)$ can be acquired by deconvolution, which is division in the frequency domain:

$$Y(f) = S_1(f)H(f) \quad (2.20)$$

$$H(f) = \frac{Y(f)}{S_1(f)} \quad (2.21)$$

$$h(t) = \mathcal{F}^{-1}(H(f)) \quad (2.22)$$

$S_1(f)$ is the Fourier transform of the synchronized swept sine $\sin(\phi_1(t))$. Its analytical form is derived in Novak et al. [12]:

$$S_1(f) = \frac{1}{2} \sqrt{\frac{L}{f}} \exp \left(j2\pi f L \left[1 - \ln \left(\frac{f}{f_a} \right) \right] - j\frac{\pi}{4} \right) \quad (2.23)$$

Using windowing, we are able to isolate each DP, $h_{m,n}(t)$, from the whole impulse response $h(t)$.

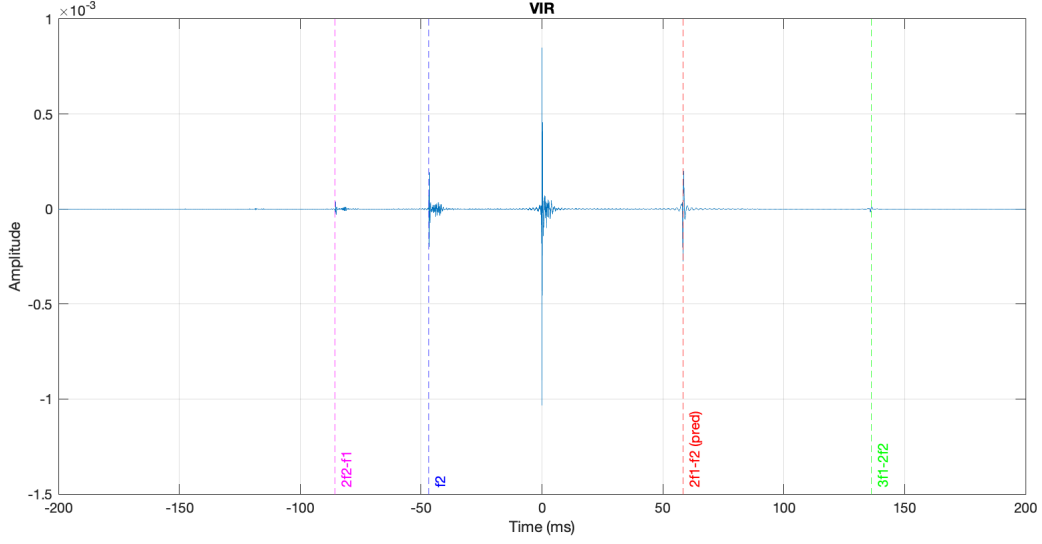


Figure 2.3: ;

The final result of deconvolution is the VIR. This consists of multiple DPs at distinct time points which are separated using temporal windowing.

Separation of SL and LL Components for DPOAEs

Earlier sections discussed that most of the important signal energy, i.e. the SL component, is located near zero time delay. The LL component has a longer delay, with most energy between 4 and 12 ms. Due to differences in phase gradient, these components can be separated from the DP-gram in the time domain using recursive exponential windows.

The whole window is constructed by adding two halves, positive and negative recursive exponential windows. For the negative half, a $\tau_c = 1$ ms is used. The positive half is constructed using a frequency-dependent τ_c determined by the formula below, derived from Moleti et. al. (2012), where $f_c[n]$ represents center frequency in Hz:

$$\tau_c[n] = \left(\frac{a}{2}\right) \left(\frac{f_c[n]}{1 \text{ kHz}}\right)^{-0.8} \quad (2.24)$$

The parameter a determines whether the window is extracting the SL component ($a =$

0.01 s) or the entire DP-gram ($a = 0.05$ s), which includes both the SL and LL components. Due to its computational efficiency, the SSS method processes the entire swept-sine response repeatedly for a multitude of windows for various $f_c[n]$. We set our center frequency equal to the instantaneous frequency of the swept sine given by:

$$f_{\text{inst}}(t) = f_a \exp\left(\frac{t}{L}\right) \quad (2.25)$$

This identifies the frequency of the sweep at time t . The number of center frequencies determines the number of windows, and therefore the number of intermodulation DPs. We can construct a matrix of intermodulation products $N \times M$ such that N is the number of center frequencies and M is the number of samples in the impulse response. Each ImIR is denoted as $\hat{h}_{DP}$:

$$\hat{h}_{DP} = (h_{DP}[1], h_{DP}[2], \dots, h_{DP}[M]) \quad (2.26)$$

$$h_{DP} = (\hat{h}_{1DP}, \hat{h}_{2DP}, \dots, \hat{h}_{NDP}) \quad (2.27)$$

Another $N \times M$ matrix, R , is constructed for the recursive exponential window r_i :

$$R = (\hat{r}_1, \hat{r}_2, \dots, \hat{r}_N) \quad (2.28)$$

The desired component, either the SL or whole DP-gram, is extracted by applying the matrix of windows to the matrix of intermodulation products via a dot product, that is, element-wise multiplication in MATLAB, and taking the Fourier transform of the end result:

$$H_{DP} = \mathcal{F}[h_{DP} \odot R] \quad (2.29)$$

H_{DP} is an $N \times M$ matrix where each row in H_{DP} symbolizes a DP-gram for a certain window $\hat{r}_N$. A pointwise artifact rejection method described by Fang and Liu [31] is applied

to each row of the matrix to disqualify outliers. After this, we calculate the column-wise mean of the matrix H_{DP} . Since the entire DP-gram consists of both SL and LL components, we can vector-subtract the SL DP-gram to find the LL component.

2.2.4 Comparisons with the LSF

The SSS and LSF methods are variations of the swept-tone method used for extracting DPOAEs. The LSF, which is a modeling technique, has served as the reference for verifying the accuracy of other emission-extraction methods since it was introduced. Thus, it is important to examine how well the SSS performs against this reference state-of-the-art model. This section compares the two methods in closer detail.

Computational Efficiency

The most significant comparative difference is in calculation time. The SSS approach is computationally much less demanding than the LSF approach. Its speed allows the measured DP-gram to be calculated and presented to the experimenter almost instantaneously. The primary factor that makes this approach so efficient is the use of frequency-domain convolution. Since convolution in the frequency domain is simply multiplication, the speed of the calculations in MATLAB is much faster.

An additional reason for the computational efficiency of the SSS is that it allows for complete system characterization, including obtaining the linear impulse response (IR) and separating the harmonic distortion responses from a single measurement; this greatly reduces the overall effort. This computational speed is why the SSS is so suitable for clinical applications. In contrast, the LSF is computationally expensive because it processes the response in time frames, which requires many individual fittings and overlapping frames to obtain good frequency resolution.

Noise Robustness and Accuracy

Among other DPOAE extraction methods, the LSF method has been shown to be the most noise-robust. Based on Vencovský et al. [13], SSS is comparable with the LSF technique in relation to noise robustness. The results yielded by the SSS technique are just as accurate as those obtained by the LSF. There is strong agreement between the DP-grams derived from both approaches.

Processing and Component Separation

Both techniques are capable of extracting the two DPOAE components, but they achieve this through different mechanisms. The LSF is a modeling technique that works by fitting an assumed DPOAE signal into the swept-sine response within a temporal frame of fixed duration. A longer analysis window, approximately 500 ms, is used to smooth the fine structure and separate the SL component.

The SSS relies on deconvolution of the output signal with the analytical expression of the input swept sine to yield a virtual impulse response. Then, a frequency-dependent window is applied instead of a fixed window. These frequency-dependent windows are capable of extracting either the entire DP-gram or the SL component, depending on the parameters used.

Despite their differences in extracting the SL component, both methods can obtain the LL component through vector subtraction of the SL from the entire DP-gram. Discrepancies in results between the two methods were larger for the reflection component than for the distortion component.

Artifact Rejection

Due to their fundamental differences in processing structure, there are also differences in the measurement of artifacts. Because the LSF uses fixed temporal windows, it allows artifacts to be discarded while keeping the rest of the response for processing, a method known as frame-

based artifact rejection. This frame-based approach is advantageous because it allows for the detection of measurement artifacts in these frames, permitting the selective mitigation of noise and artifacts. This feature is what gives the LSF its noise-robust qualities. In contrast, SSS processes the entire response, which may take several seconds. This is one of the few situations in which the LSF clearly outperforms the SSS.

2.3 Parameter Analysis and Optimization

Although the synchronized swept-sine technique is computationally more efficient, it does not yet provide superior results. The overarching goal of this study is to find optimal parameters for component separation and extraction that yield the highest-quality results.

By identifying optimal parameters, we aim to maximize the potential of the SSS technique. Perfect separation is not attainable, but we aim to maximize the temporal distance between the SL and LL components of each DP. However, the quality of the extracted SL component is crucial. Maximizing the temporal distance between components is not useful if the quality of the extracted distortion component is low.

2.3.1 Parameters for Temporal Separation

The SSS technique relies on the time-domain characteristics of the components. One aim of this study is to optimize the two factors that govern temporal separation: sweep rate and the frequency ratio α .

Sweep Rate

Sweep rate is defined as the instantaneous slope of the frequency-versus-time curve and dictates how fast the stimulus frequency changes as it travels from start frequency f_{1a} to stop frequency f_{1b} . The two parameters pertaining to sweep rate are the sweep rate itself, r , measured in octaves per second, and the sweep-rate coefficient L , measured in seconds, which

describes the rate of exponential increase in frequency. These parameters are mathematically related and can be expressed as an inverse relationship:

$$L = \frac{1}{r \ln(2)} \quad (2.30)$$

From the sweep-rate coefficient, we can calculate the duration of the sweep, T , in seconds. Returning to Equation (3), we solve for T and obtain:

$$T = L \ln \left(\frac{f_{1b}}{f_{1a}} \right) \quad (2.31)$$

Using a lower T implies decreasing the total duration of a clinical acquisition trial, which increases the efficiency of the process. This can be achieved by increasing the sweep rate r .

However, r must be increased within reason. If r is set too high, the separation between DPs decreases, causing overlap between them. Furthermore, faster sweeps have a tendency to cause slight phase shifts between the SL and LL components, which lowers the accuracy of the results. If r is set too low, T becomes too high, which means a longer data-acquisition period. Also, the components become overly separated, meaning there is unnecessary empty space between them. Vencovský et al. [13] showed that using a sweep rate of 0.5 oct/s provides a time difference of 644 ms between impulse responses. A sweep rate of 10 oct/s is considered a theoretical upper limit, since a higher rate would likely lead to overlap between components, whereas a sweep rate of 5 oct/s is considered a practical upper limit because it is more clinically viable and guarantees clear separation of intermodulation products.

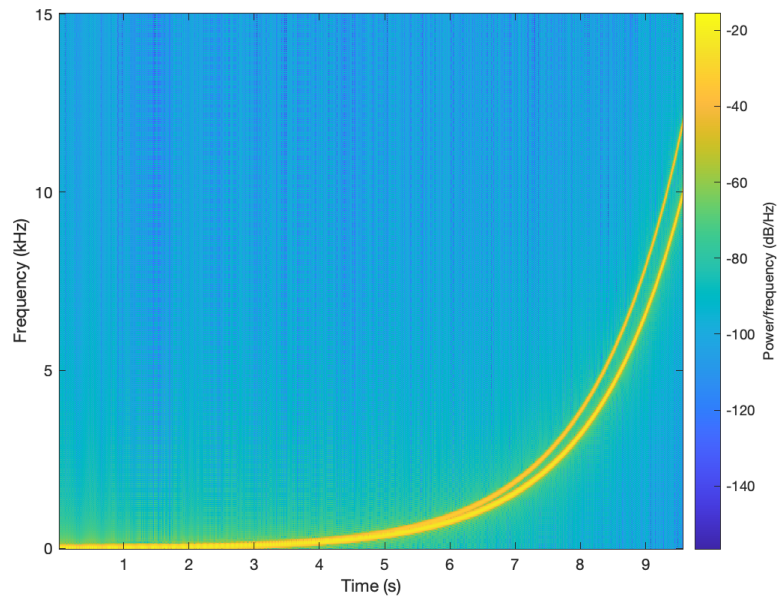


Figure 2.4: A spectrogram example of a low sweep rate stimulus ($r = 0.5$ oct/s).

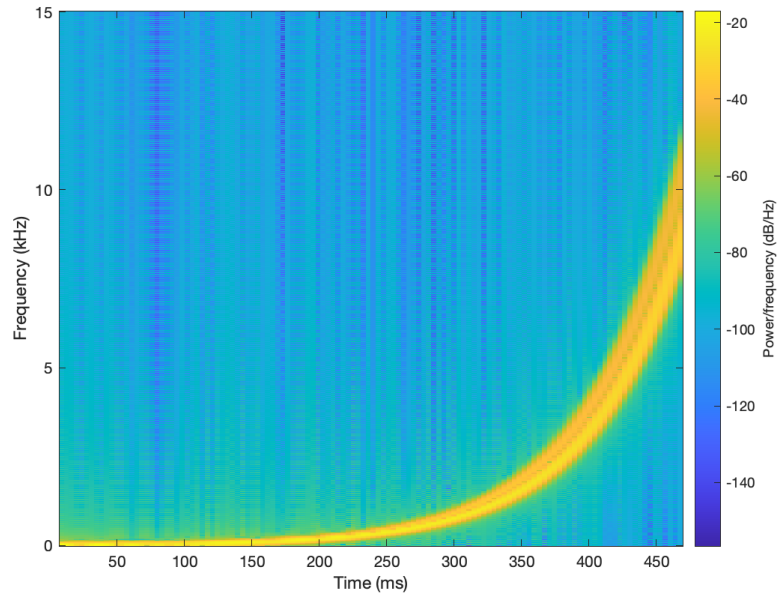


Figure 2.5: A spectrogram example of a higher sweep rate stimulus ($r = 10$ oct/s). The duration of the sweep is substantially shorter but has noticeably worse resolution.

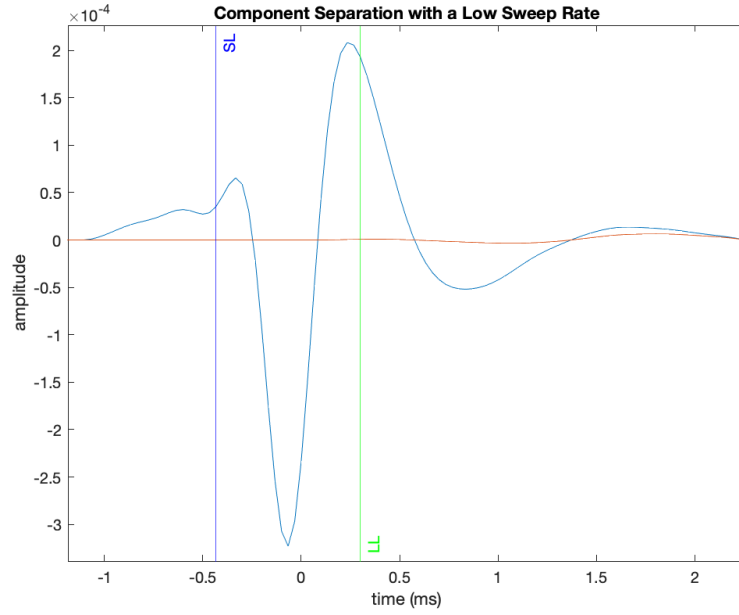


Figure 2.6: An example of the extracted components using a low sweep rate ($r = 0.5$ oct/s). The lines represent the approximate onsets of the SL and LL components respectively.

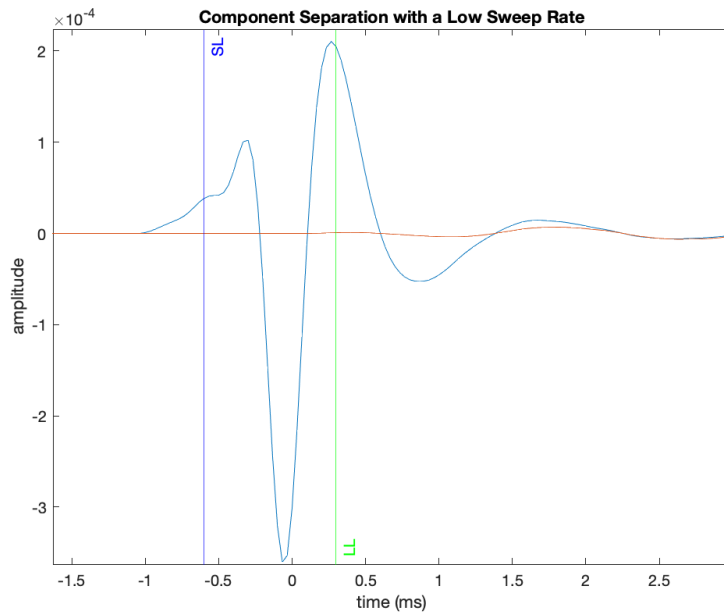


Figure 2.7: An example of the extracted components using a high sweep rate ($r = 10$ oct/s).

Frequency Ratio

α is defined as the frequency ratio f_2/f_1 . Based on Equation (13), we observe that increasing α increases time delay, which this study aims to maximize. By increasing time delay, more effective temporal windowing becomes possible, which is critical for SL and LL component separation.

Similar to sweep rate, α must be increased within practical limits. Maximizing α is constrained by the physical mechanisms of the cochlea. If α is increased too much, the physical separation on the basilar membrane between the stimulus tones becomes too great. As a result, the measurement will have reduced amplitude and, consequently, a lower signal-to-noise ratio (SNR). If α is too low, the stimulus tones may mutually interfere via two-tone suppression. This will also reduce amplitude and SNR. Therefore, the optimization problem is how to maximize α without sacrificing measurement quality.

Both Vencovský et al. [6] and Durante et al. [5] attempted to identify an optimal range for α . Vencovsky's results suggested a range from 1.05 to 1.4, which is quite broad. Durante's results suggested a narrower range from 1.18 to 1.28. It should be noted that both Vencovsky and Durante used the LSF method rather than the SSS technique. More recently, Glavin et al. [32] and Vencovský et al. [13] proposed optimal values of 1.22 and 1.2, respectively.

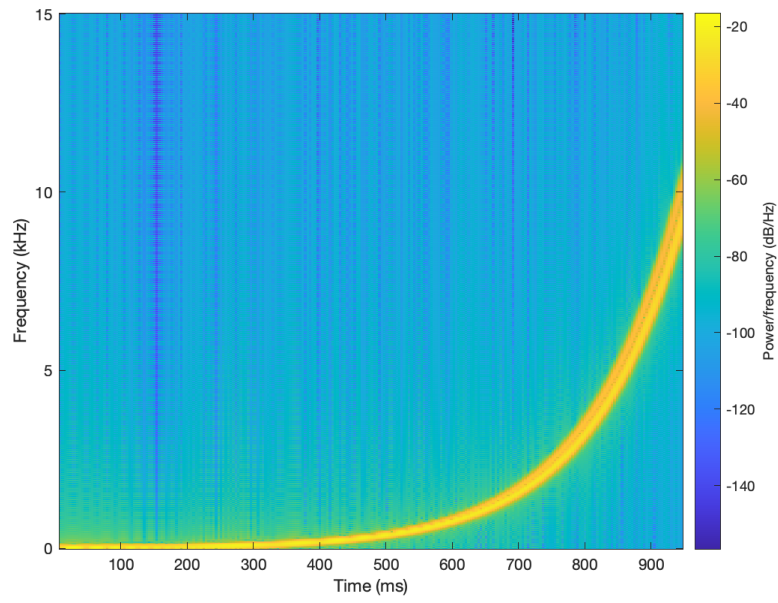


Figure 2.8: SSS stimulus generated using $\alpha = 1.1$.

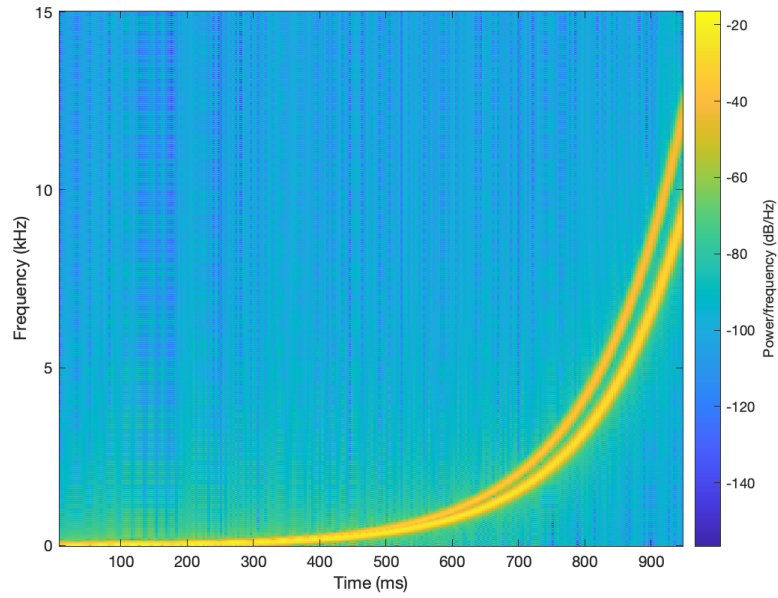


Figure 2.9: SSS stimulus generated using $\alpha = 1.3$. Increasing α means greater space between the f_1 and f_2 waves.

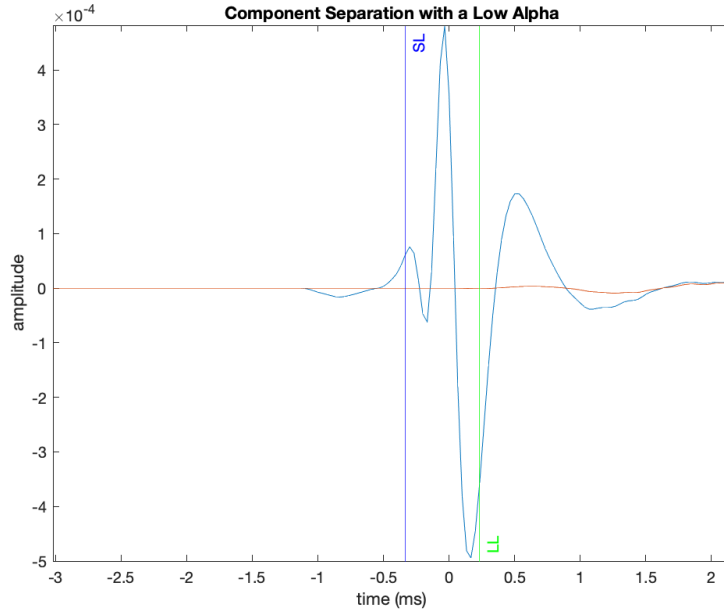


Figure 2.10: Extracted SL and LL components from a DPOAE generated using $\alpha = 1.1$.

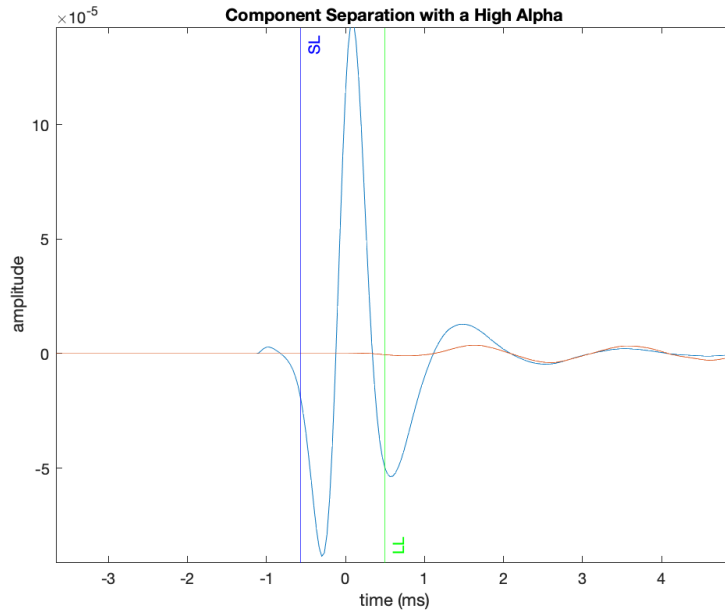


Figure 2.11: Extracted SL and LL components from a DPOAE generated using $\alpha = 1.3$. A higher alpha value shows greater temporal separation between components, but also lower amplitude.

2.3.2 Parameters for Amplitude

When analyzing the quality of the extracted SL component, we reference the amplitude of the distortion. A higher-amplitude signal means a higher SNR and, hence, a better-quality measurement. The parameters most related to amplitude are the stimulus levels L_1 and L_2 for the f_1 and f_2 waves, respectively.

Zelle et al. [33] showed that for a given L_2 , there exists a unique optimal L_1 that evokes maximum amplitude. Optimizing these levels involves balancing the mechanical properties of the cochlea with the perceptual characteristics of human hearing. DPOAEs are generated in the region where the traveling waves of the two primary tones overlap maximally, which is close to the tonotopic place of the higher-frequency tone, f_2 . For the cochlea to produce the largest possible emission, the two traveling waves must have approximately equal mechanical amplitudes at the f_2 tonotopic place. The challenge in achieving this balance stems from the nonlinear nature of the cochlear amplifier, which compresses the magnitude of vibrations at different rates depending on where the wave is relative to its peak. At the f_2 tonotopic place, the f_2 traveling wave is at its maximum amplitude. In this region, the cochlear amplifier is highly active, and the response is strongly compressed. Because f_1 is a lower frequency, its tonotopic peak is located further apically. When it passes through the f_2 site, it is still on its basal tail. In this tail region, the f_1 wave is relatively linear and less compressed [34]. Since the f_2 wave is heavily compressed and the f_1 wave is not, their respective stimulus levels, L_1 and L_2 , must be set so that they result in equal vibrations at the interaction site. If L_1 is too low, the f_1 wave is too weak at the f_2 site to produce a strong distortion. If L_1 is too high, the f_1 wave suppresses the f_2 wave, resulting in DPOAE amplitude reduction.

To compensate for the high compression of the f_2 wave, researchers traditionally set L_1 higher than L_2 [35]. It is also necessary for L_1 to be greater than L_2 because higher frequencies are perceived as louder than lower frequencies. Thus, the lower f_1 wave requires a higher stimulus level to achieve the same perceptual loudness, which ensures that each

frequency within the sweep is presented at a perceptually consistent level. Studies also show that the optimal difference between L_1 and L_2 decreases as L_2 increases Lijun and Chen [36] and approximately results in a linear relationship between L_1 and L_2 [37]. This linear relationship can be expressed using the scissor paradigm. The scissor paradigm was introduced by Kummer et al. [38] and is designed to mathematically adjust stimulus levels to account for the disparate compression rates. The mathematical relation is described below:

$$L_1 = 0.4L_2 + 39 \text{ dB} \quad (2.32)$$

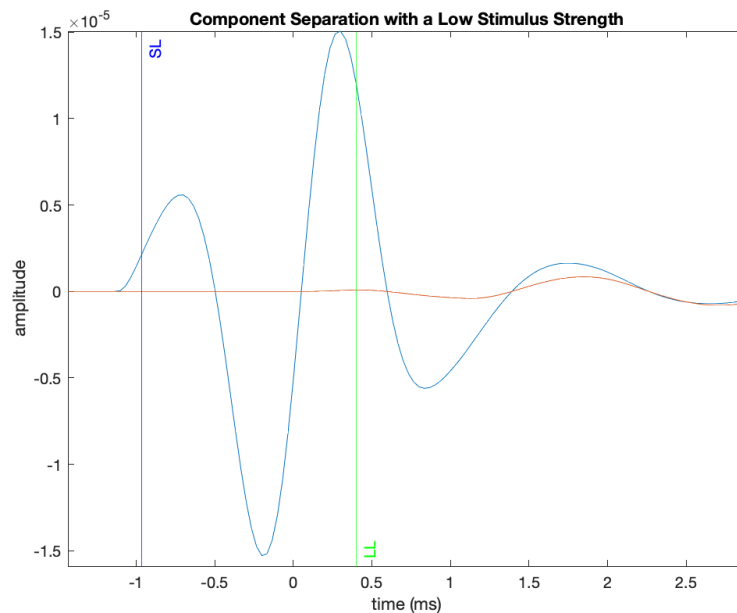


Figure 2.12: An example of the extracted components using a low stimulus level of $L_1 = 45dB$ and $L_2 = 40dB$.

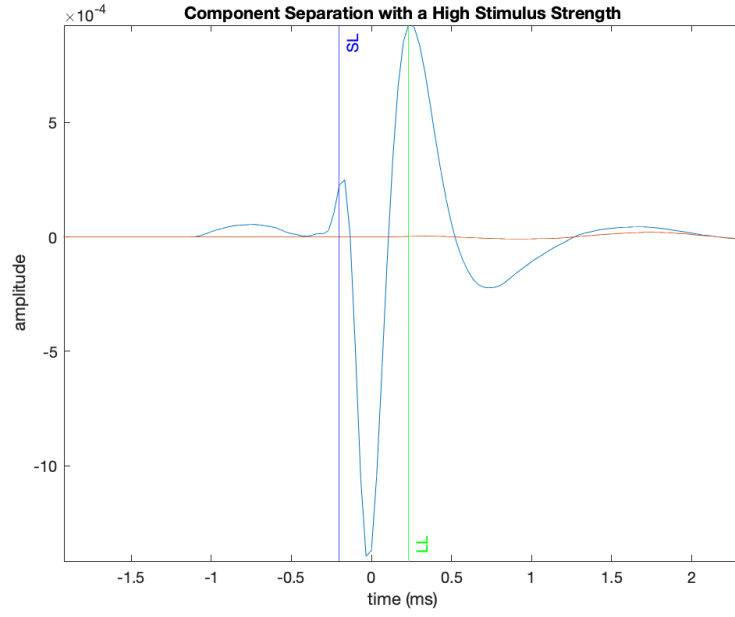


Figure 2.13: An example of the extracted components using a high stimulus level of $L_1 = 70dB$ and $L_2 = 60dB$. The amplitude for the SL component is noticeably greater in this case.

Chapter 3

Specific Aims

The objective of this study is to identify optimal parameters for the SSS technique for component separation and extraction of DPOAEs. At this point in time, it is equivalent to the current state-of-the-art method, LSF. By optimizing the parameters, we can truly maximize the potential of the SSS technique. The parameters that provide the maximum amount of latency between the SL and LL components, as well as maximum SL amplitude, or the best tradeoff between the two, will be considered optimal.

3.1 Outcomes

To measure the success or failure of a parameter, we will use three objective outcomes. These metrics are as follows:

1. Separation between the SL and LL components
2. Percentage of overlap between the SL and LL components
3. SNR of the SL component

The separation metric is designed to measure the temporal spacing between the approximate onset of the SL and LL components. First, the peak of both components is identified.

Using this identified peak, we calculate the threshold as 10% of peak amplitude. The index at which the signal crosses this threshold is determined to be the onset. By plugging this index into the time vector, we find the corresponding time, in seconds, for each component. Lastly, the time difference between the two is calculated via subtraction and converted to milliseconds:

$$\text{Separation (ms)} = (LL_{\text{onset}}(\text{s}) - SL_{\text{onset}}(\text{s})) \cdot 1000 \quad (3.1)$$

Higher separation indicates better temporal spacing between the SL and LL components, and vice versa.

The percentage of overlap aims to quantify how much temporal overlap exists between the SL and LL signals within the time-frame in which the SL signal is most dominant. For the time-frame starting point, we can use the approximate SL onset determined for the separation metric. The SL component has an approximate duration of 2 ms; using this reference, we define the index on the time vector corresponding to 2 ms and set that as the time-frame endpoint. First, we identify how many samples each signal has in the active time-frame of the SL component. The ratio of the two is then taken and converted to a percentage:

$$SL_{\text{total samples}} = SL_{\text{end index}} - SL_{\text{onset index}} \quad (3.2)$$

$$LL_{\text{samples}} = SL_{\text{end index}} - LL_{\text{onset index}} \quad (3.3)$$

$$\text{Overlap \%} = \left(\frac{LL_{\text{samples}}}{SL_{\text{total samples}}} \right) \cdot 100 \quad (3.4)$$

A lower overlap indicates better separation between the DP components; in other words, it is inversely related to separation.

The SNR of the SL component quantifies the quality of extraction of the SL component, that is, how much amplitude, or SL signal, relative to noise or residue, lies within the designated timeframe. Using the same indices as in the overlap calculations, we take the magnitude of the SL signal and then calculate the root-mean-square (rms) of that magnitude. The rms of the tail end of the DP itself functions as a noise-floor estimate. The ratio of rms amplitudes is then converted to decibels using the formula below:

$$\text{SNR}_{SL}(\text{dB}) = 20 \log_{10} \left(\frac{\text{signal}_{\text{rms}}}{\text{noise}_{\text{rms}}} \right) \quad (3.5)$$

$$\text{signal}_{\text{rms}} = \text{rms}(|SL_{\text{signal}}|) \quad (3.6)$$

$$SL_{\text{signal}} = [SL_{\text{start}}, \dots, SL_{\text{end}}] \quad (3.7)$$

$$\text{noise}_{\text{rms}} = \text{rms}(h_{m,n}(t)(\text{end}/2 : \text{end})) \quad (3.8)$$

The rms of a signal x is defined below:

$$\text{rms}(x) = \sqrt{\frac{1}{n} \sum x_i^2} \quad (3.9)$$

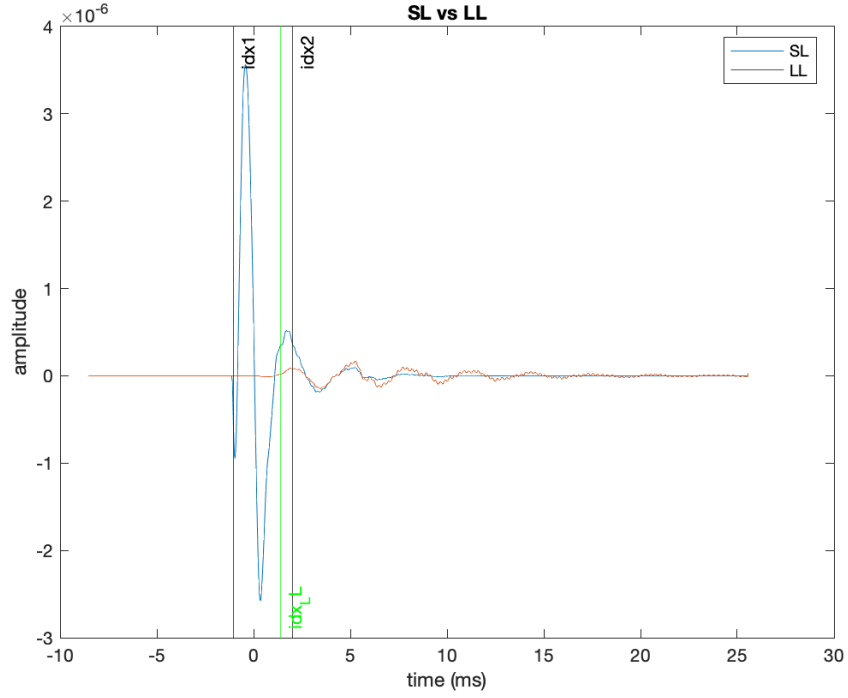


Figure 3.1: A visual demonstration of what low overlap (higher temporal separation) looks like. This figure features the $3f_1 - 2f_2$ DP with an approximate overlap of 20%. This was obtained using $r = 2.5$ oct/s, $\alpha = 1.27$, and $L_1 = 60dB$ and $L_2 = 50dB$. Idx 1 represents the estimated onset of the SL component and idx 2 is the two ms mark (approximate duration of the SL component). The LL idx marks the estimated onset of the LL component. The closer the LL onset is to the SL onset, the higher the overlap.

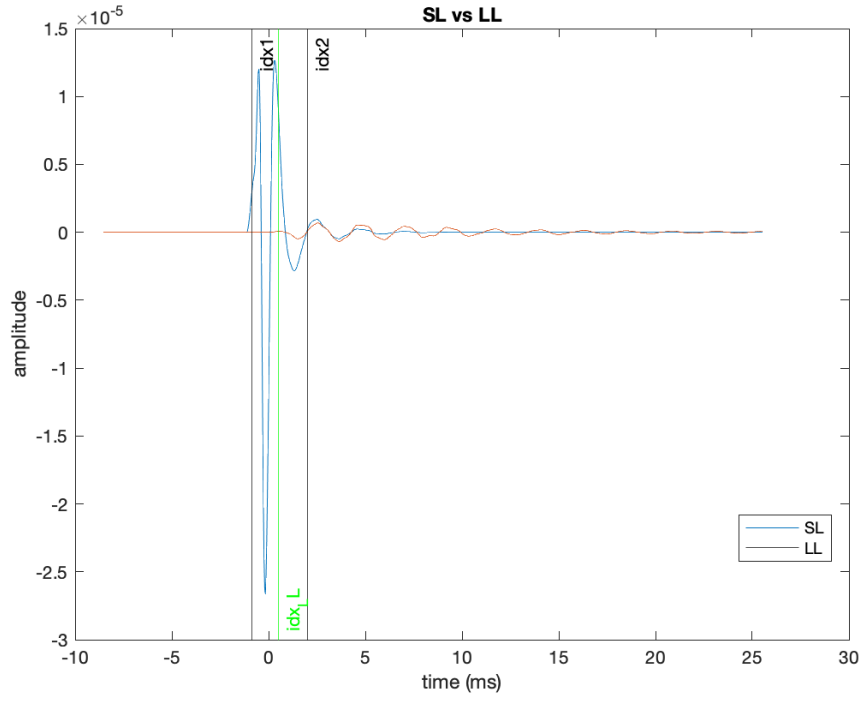


Figure 3.2: A visual demonstration of what medium overlap (medium temporal separation) looks like. This figure features the $3f_1 - 2f_2$ DP with an approximate overlap of 50%. This was obtained using an $r = 0.5$ oct/s, $\alpha = 1.18$, $L_1 = 60dB$ and $L_2 = 50dB$.

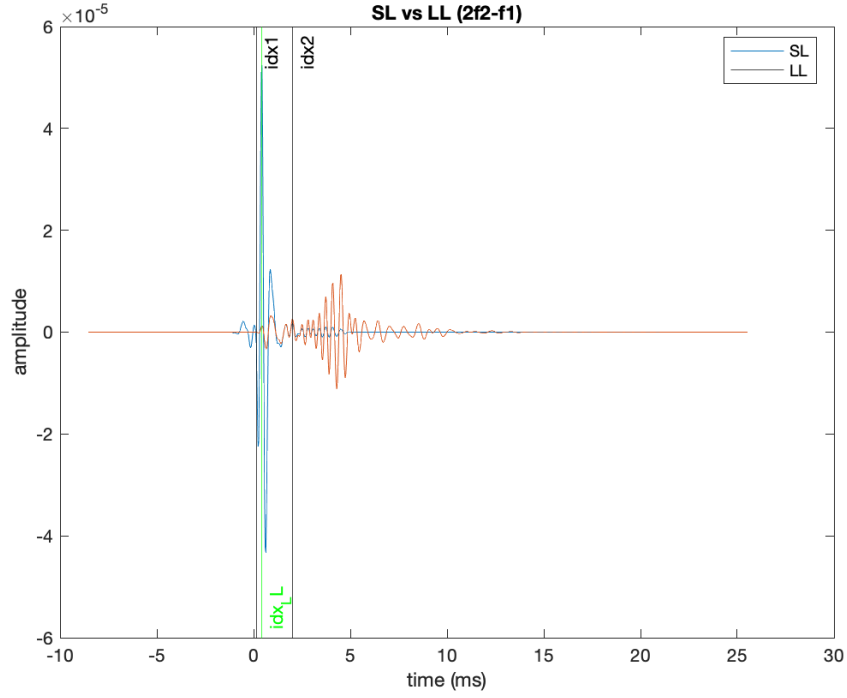


Figure 3.3: A visual demonstration of what high overlap (low temporal separation) looks like. This figure features the $2f_2 - f_1$ DP with an approximate overlap of 85%. This was obtained using an $r = 0.5$ oct/s, $\alpha = 1.2$, $L_1 = 60dB$ and $L_2 = 50dB$.

3.2 Hypothesis

Based on previous research and what we know about cochlear mechanics, we can predict how the metrics will behave as we iterate through different parameter values.

3.2.1 Sweep Rate

Of all the parameters, I expect sweep rate to have the least impact on the metrics. Component separation depends on where the distortion is generated in relation to the characteristic place of the DP frequencies. The speed of the stimulus does not influence this. From Venkovský et al. [13], we can observe that increasing the sweep rate compresses the different

DPs closer together in time. Because of this, it is possible, that the SL and LL components of the different DPs will exhibit less temporal distance.

However, this study will not test sweep rates above Novak’s practical limit of 5 oct/s, which suggests that the components remain well separated up to this point. Thus, I expect sweep rate to have little to no effect on overlap as well. Based on previous research by Shera [23], the choice of sweep rate does not significantly alter the actual biological emission, that is, the amplitude, generated by the cochlea. I expect the simulations to corroborate this. There will be no significant relationship between the SNR of the SL component and sweep rate.

3.2.2 Alpha (α)

We defined α as the frequency ratio between the f_2 and f_1 waves, that is, $\alpha = f_2/f_1$. If we were to unwind the basilar membrane (BM), we could represent it as a line such that one end corresponds to the base and the other to the apex. Along the BM lies the characteristic place for each audible frequency: higher frequencies resonate at the basal end, whereas lower frequencies resonate at the apical end.

Since α dictates the spacing between the f_1 and f_2 waves, it also dictates the characteristic places of these frequency waves. As α increases, the spacing increases and vice versa. In other words, the f_2 wave moves closer to the base while the f_1 wave remains stationary. The distortion component is wave-fixed for a specific α and is generated in the region of maximum overlap between the f_1 and f_2 waves, generally closer to the f_2 wave.

However, α also dictates the characteristic place of the different DPs. The kind of effect α has on the DP depends on the frequency of the DP. For the DPs below the f_2 wave, that is, $2f_1 - f_2$ and $3f_1 - 2f_2$, increasing α decreases the distortion frequency, meaning the characteristic place shifts apically, and vice versa. Thus, the reflected energy has to travel more apically, which increases the separation between the SL and LL components.

For DPs above the f_2 wave, that is, $2f_2 - f_1$, increasing α increases the distortion fre-

quency, meaning the energy must travel basally to be reflected, and vice versa. This would cause the latency between the SL and LL components to decrease. In essence, when α is increased, the DP frequencies move away from the f_1 wave. When α is decreased, the DP frequencies move toward the f_1 wave.

Between the three parameters, I expect α to have the greatest effect on component separation and, consequently, on overlap as well. A higher latency means there is more temporal separation between components, and thus they are less likely to bleed into one another. With respect to SNR, I expect the optimal α to yield the highest SNR. If α is set too low, there is more overlap between the f_1 and f_2 waves. Due to their proximity, the waves suppress each other, resulting in lower amplitude. If α is set too high, the two waves do not interact enough and produce a weaker distortion with lower amplitude.

3.2.3 Stimulus Strength

The strength of the stimulus will have little to no effect on separation or overlap, but it will have a significant effect on amplitude. Stimulus level affects the amplitude of the different components differently. For the SL component, amplitude reacts differently depending on the stimulus level. At low stimulus levels, approximately 20–50 dB, the amplitude increases linearly with stimulus level. A 1 dB increase in stimulus corresponds to a 1 dB increase in amplitude. At moderate stimulus levels, approximately 50–60 dB, the amplitude increases at a slower rate through compressive growth. A 1 dB increase in stimulus level corresponds to an approximate increase of 0.2–0.6 dB in amplitude. At high stimulus levels, approximately 60 dB and above, there is no amplitude increase [39]. In contrast, the LL component shows compressive growth across all stimulus levels [9]. Therefore, the optimal stimulus is expected to exist around moderate-strength levels because the SL component will benefit from increased amplitude from higher stimulus levels without too much interference from the compressive growth of the LL component.

Chapter 4

Methods

Data will be collected using simulations. Each tested parameter will have its own simulation in which the tested parameter iterates through an array of values while the other parameters are constant. Each parameter simulation will be run twice: one simulation without noise and one with additive white Gaussian noise (AWGN). Some parameters that will be held constant across all simulations are the sample rate (30 kHz), minimum frequency of the f_1 wave (700 Hz), maximum frequency of the f_1 wave (9 kHz), and the number of simulations run for each tested parameter (10). The frequency of the f_2 wave is calculated by multiplying the f_1 wave by α .

4.1 Identifying Test Ranges for Parameters

Before any analysis, we need to solidify the range of test values. This subsection goes over the selected range for each parameter and how this range was decided.

4.1.1 Sweep Rate

Previously, a practical upper limit on sweep rate of 5.0 oct/s was specified. Both Abdala and Luo [10] and Vencovský et al. [13] suggested that 0.5 oct/s was optimal for component

separation and extraction. Relative to the practical and theoretical upper limits, a sweep rate of 0.5 oct/s can be considered low. A lower sweep rate requires more time to sweep through the frequency range, making the measurements less time-efficient. Thus, we can use 0.5 oct/s and 5.0 oct/s as the lower and upper limits for this study. We iterate through this range in increments of 0.5.

4.1.2 Alpha

Based on research and observations from Durante et al. [5], an approximate optimal range for α is 1.18–1.28, although these values were determined using the LSF technique. The test range for alpha in the simulations will be identical to Durante’s estimated range.

4.1.3 Stimulus Strength

In current research, it is common for researchers to use certain predetermined sets of stimulus levels. Among the most common in recent studies are:

$$L_1 = 70 \text{ dB}, \quad L_2 = 60 \text{ dB} \quad [32]$$

$$L_1 = 65 \text{ dB}, \quad L_2 = 55 \text{ dB} \quad [5]$$

$$L_1 = 60 \text{ dB}, \quad L_2 = 50 \text{ dB} \quad [13]$$

$$L_1 = 50 \text{ dB}, \quad L_2 = 45 \text{ dB} \quad [13]$$

$$L_1 = 45 \text{ dB}, \quad L_2 = 40 \text{ dB} \quad [33]$$

Another method for choosing appropriate stimulus levels is using the scissor paradigm as described in Section 4.2. Using the common L_2 values from above, we can apply Equation (2.32) to obtain additional test values:

$$L_1 = 63 \text{ dB}, \quad L_2 = 60 \text{ dB}$$

$$L_1 = 61 \text{ dB}, \quad L_2 = 55 \text{ dB}$$

$$L_1 = 59 \text{ dB}, \quad L_2 = 50 \text{ dB}$$

$$L_1 = 57 \text{ dB}, \quad L_2 = 45 \text{ dB}$$

$$L_1 = 55 \text{ dB}, \quad L_2 = 40 \text{ dB}$$

The combination of these two sets will constitute the test values for stimulus strength.

4.2 Simulation Details

Simulations will be conducted in MATLAB using a simulated cochlear response. The goal is to simulate the SSS technique as implemented by [13]. For programming-efficiency purposes, each parameter we are optimizing, α , sweep rate, and stimulus strength, will be tested in a separate script. All three parameter simulations follow a similar process, which can be broken down into two phases:

1. Preprocessing
2. Evaluation

There is also a diagram at the end of the section to visually demonstrate the two phases of the simulation workflow.

4.2.1 Preprocessing

This is the data-generation phase. Prior to the actual simulations, the data are preprocessed using `verhulst2012.m` from the `amtoolbox` (Auditory Modeling Toolbox) to simulate the

cochlear response. This toolbox can be downloaded in MATLAB as a package. After initializing the constants and the cell array where the generated OAEs will be stored, we iterate through the array of test parameter values. For each iterated value, we create 10 synchronized swept sines, one for each simulation. Each synchronized swept sine is inserted into the simulated cochlear model and produces a unique OAE response. The final output is saved into the MATLAB workspace and can be accessed at any time.

4.2.2 Evaluation

This is the simulation phase. Similarly to the preprocessing phase, we start by declaring initial parameters and looping through the array of parameter test values. However, this time we call the OAE from the preprocessed data set corresponding to the test value and simulation number and apply deconvolution to obtain our virtual impulse response (VIR). From this VIR, we are able to separate the different DPs based on their different latencies and then separate their individual components using a recursive exponential window.

The metrics are then computed for each DP as described in Section 3.1 and saved as a structure array. After the simulation data are acquired, the data are plotted and organized into a `.csv` file for statistical analysis. A two-way analysis of variance (ANOVA) is implemented in MATLAB for each metric to evaluate the main effects of the test parameter, DP type, and any possible interaction between the two. The conclusion of the statistical analysis marks the conclusion of the simulations. In noisy simulations, AWGN is added to the simulated OAE prior to deconvolution, and the process is identical afterward.

Simulation Workflow

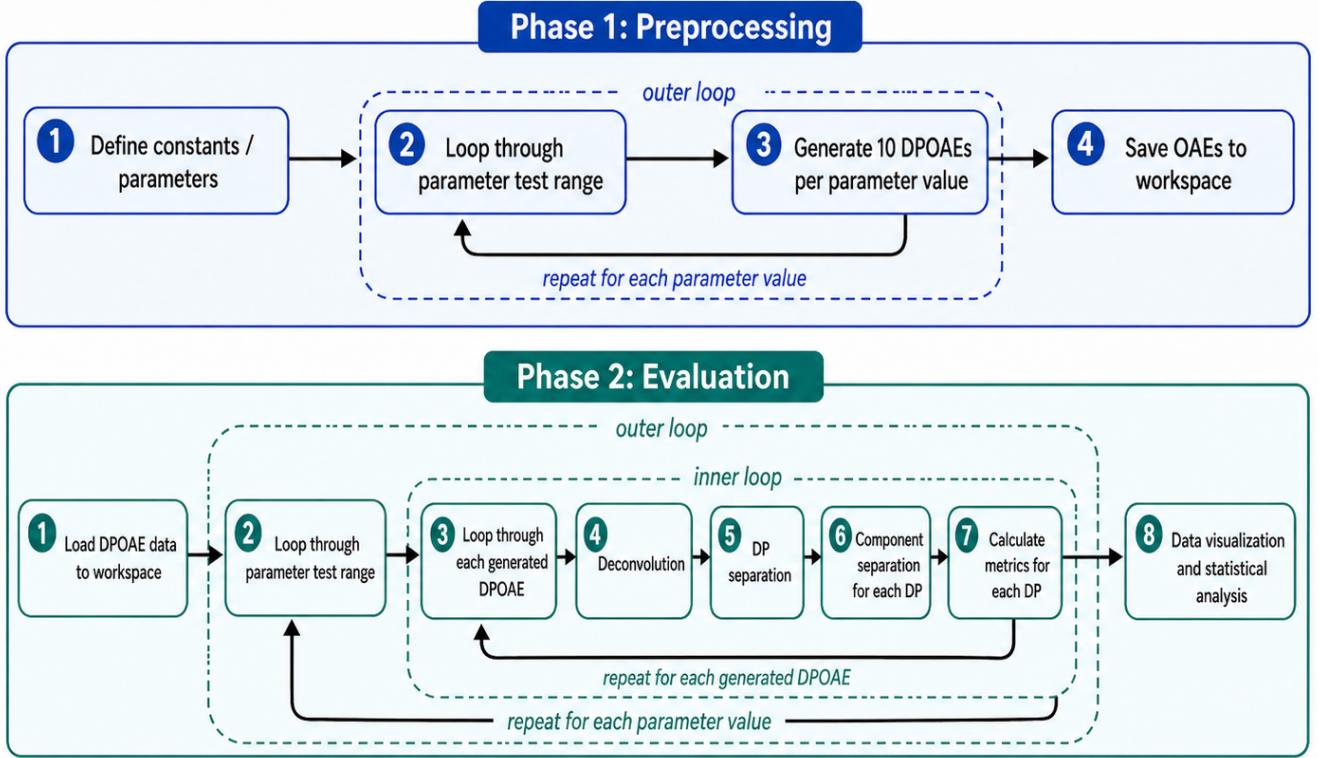


Figure 4.1: A diagram of the preprocessing and evaluation phases of the simulations.

4.3 Important Notes

For application of the SSS technique, the simulations use custom functions made by Antonín Novák to generate the synchronized sweeps (`synchronized_swept_sine.m`), create the analytic signal used for deconvolution (`synchronized_swept_sine_spectra.m`), and separate the DPs (`synchronized_swept_sine_IR_separation.m`). These functions are publicly available and can be found on Novák's website:

To account for the predicted negative latency of the $2f_2 - f_1$ DP, some adjustments were made to Novák’s IR separation function. In the original function, a dynamic upper bound is used to ensure that each extracted IR stays within valid signal limits and avoids overlap with previously extracted IRs. A signal with negative latency would be outside those limits. By removing this upper bound, the updated version instead treats the IR as circular, meaning the extraction can wrap around the end of the IR, allowing us to recover the $2f_2 - f_1$ DP. However, this means that we have to index the $2f_2 - f_1$ differently from the others. This is done by adding one sweep period T to the expected latency, enabling us to represent a negative latency in a periodic fashion.

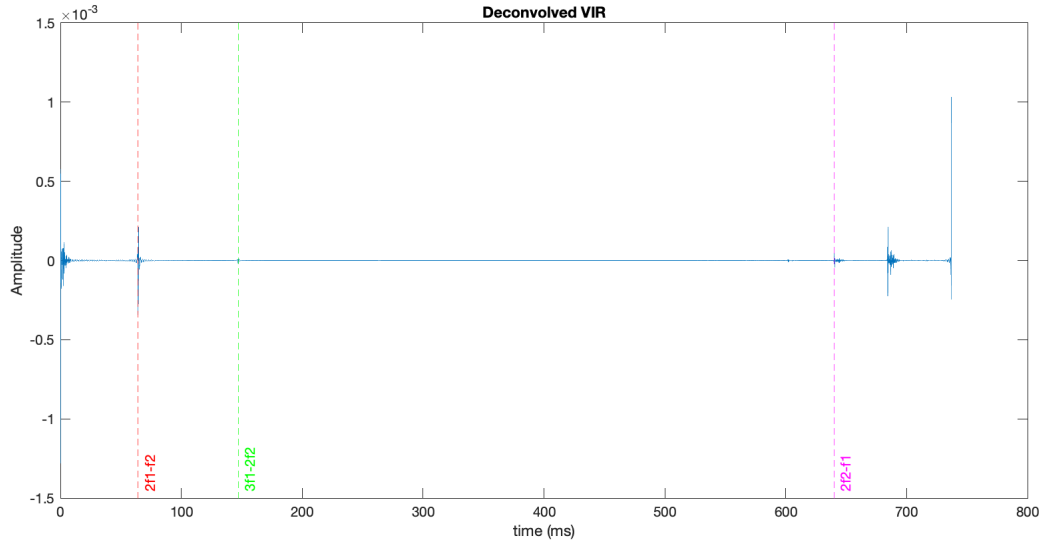


Figure 4.2: A visual demonstration of how the VIR is handled circularly to help extract the $2f_2 - f_1$ DP.

Chapter 5

Results

When interpreting the simulation results, the overall pattern of behavior is more meaningful than the exact numerical values themselves. The purpose of the simulations is not to claim that a specific output value will occur precisely in practice, but rather to reveal how the system responds as key parameters are varied. For that reason, the consistent direction, shape, and relative differences in the trends carry more interpretive value than any single numerical result.

The main effect of DP type is highly significant across all metrics and simulations, which was expected and accurately reflects cochlear behavior. For temporal separation and overlap, lower-frequency distortion products will always have greater separation and, therefore, less overlap. This is because the reflection component of lower-frequency components must travel farther apically to reach its characteristic place before it can reflect back and interact with the distortion energy, and vice versa. Since $3f_1 - 2f_2$ is the lowest-frequency distortion, it will always have the highest temporal separation. The $2f_2 - f_1$ is the highest-frequency distortion and will have the lowest temporal separation. The $2f_1 - f_2$ lies between $3f_1 - 2f_2$ and $2f_2 - f_1$ with respect to frequency, so it will usually have intermediate separation.

In terms of amplitude, the $2f_1 - f_2$ is the strongest and $3f_1 - 2f_2$ is the weakest. This directly translates to the $2f_1 - f_2$ invariably having the highest SNR and $3f_1 - 2f_2$ the lowest

SNR, with $2f_2 - f_1$ in between.

For the simulations without noise, we do not report SNR. This is because there is no real noise, so the resulting values are inflated and not meaningful. To see the ANOVA results, please refer to Appendix A.

5.1 Sweep Rate Results

In simulations without noise, latency and overlap showed little change with respect to sweep rate. In simulations with noise, all metrics showed more variability and some signs of a trend. Higher sweep rates were associated with greater latency and less overlap, but also with lower SNR.

5.1.1 Temporal Separation

The ANOVA results indicate a statistically significant main effect of sweep rate on latency, as well as a significant interaction between sweep rate and DP type. However, the plots suggest that the magnitude of this effect is relatively small and not equally pronounced across all DP types. In the non-AWGN condition, $3f_1 - 2f_2$ and $2f_2 - f_1$ showed little to no consistent change in latency across sweep rates, while $2f_1 - f_2$ showed only a slight increase. Thus, although the statistical test detected significance, the practical effect appears modest and should be interpreted cautiously.

Under noisy conditions, separation increased overall for all DP types. This may reflect a genuine noise-related effect, although it is also possible that noise influenced the threshold-based calculation procedure and artificially inflated the measured latency values. In this condition, $3f_1 - 2f_2$ showed greater variability, but no strong trend was observed. By contrast, $2f_1 - f_2$ and $2f_2 - f_1$ showed a slight increase in latency with increasing sweep rate, and the statistical analysis suggests that these increases are significant.

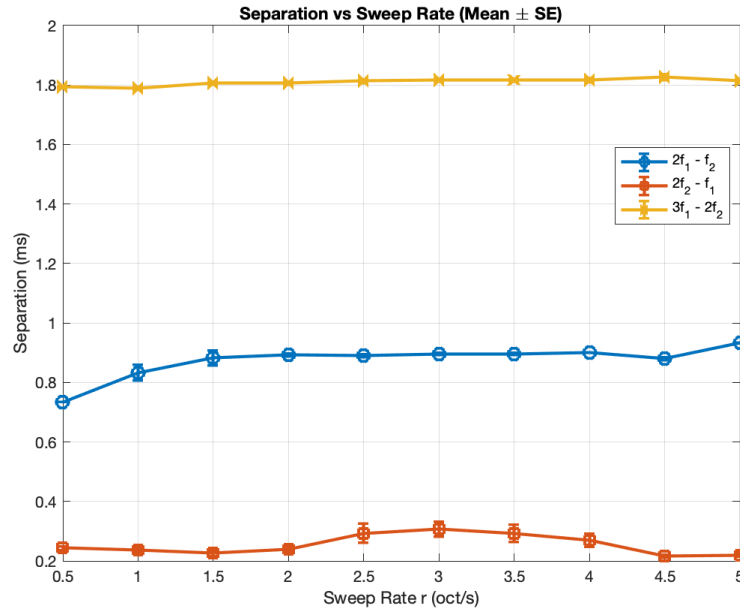


Figure 5.1: Simulation results for Sweep Rate vs. Temporal Separation of Components (no AWGN).

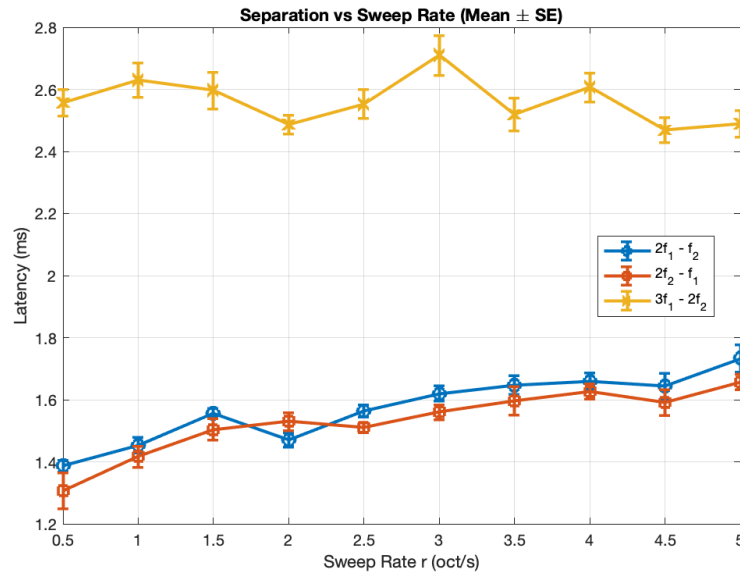


Figure 5.2: Simulation results for Sweep Rate vs. Temporal Separation of Components (with AWGN).

5.1.2 Overlap

The ANOVA results for overlap indicate statistically significant main effects of both sweep rate and DP type, as well as a significant interaction between the two factors. Across conditions, $2f_2 - f_1$ consistently showed the highest overlap percentages, $3f_1 - 2f_2$ consistently showed the lowest, and $2f_1 - f_2$ remained intermediate. Overall, this pattern suggests that distortion products with greater temporal separation tend to exhibit lower overlap, which is consistent with the expectation that components that are farther apart in time interfere less with one another.

For simulations without noise, overlap showed little variation as sweep rate increased, with only a slight decrease visible for $2f_1 - f_2$. While the statistical analysis identified sweep rate as significant, the practical extent of this effect appears relatively small in the noiseless condition and should therefore be interpreted carefully. Under noisy conditions, overlap decreased more clearly for both $2f_1 - f_2$ and $2f_2 - f_1$, which may reflect the same mechanism that produced increased latency in these distortion products. For $3f_1 - 2f_2$, overlap exhibited greater variability in the presence of noise, but no strong trend was evident. Thus, although the statistical tests support a significant effect of sweep rate on overlap, the visual patterns suggest that this effect is modest overall.

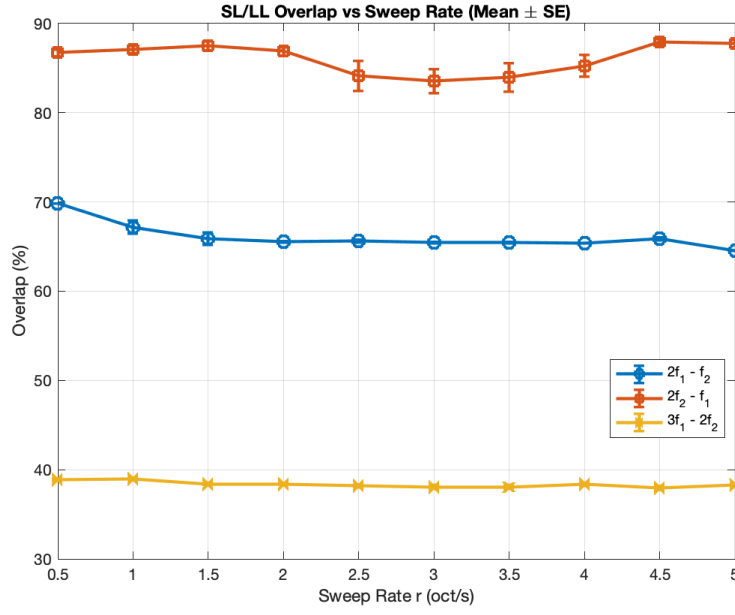


Figure 5.3: Simulation results for Sweep Rate vs. Overlap % of Components (no AWGN).

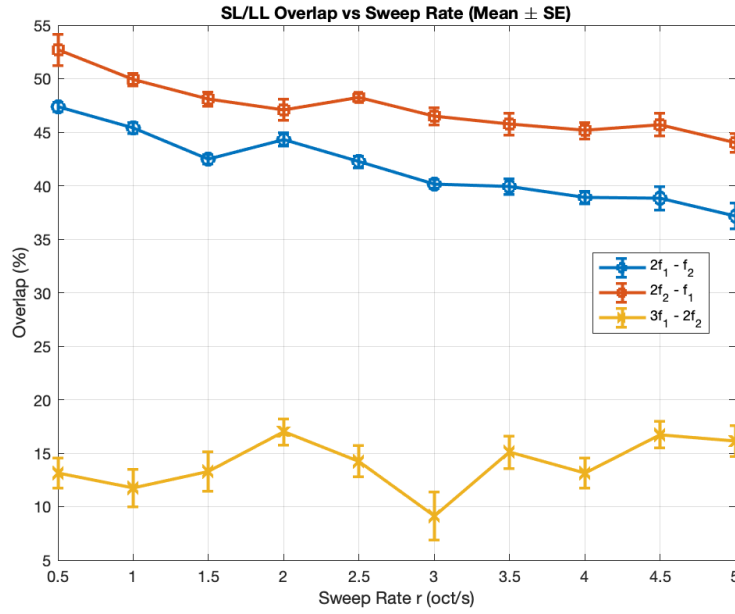


Figure 5.4: Simulation results for Sweep Rate vs. Overlap % of Components (with AWGN).

5.1.3 SNR

For the SNR of the SL component, an F-statistic of 487.6 indicates a strong effect of sweep rate. As sweep rate increased, SNR generally decreased. Across all three DPs, the lowest sweep rate, $r = 0.5$ oct/s, produced the highest SNR. For $2f_1 - f_2$ and $3f_1 - 2f_2$, the highest sweep rate, $r = 5$ oct/s, gave the lowest SNR, whereas the minimum SNR for $2f_2 - f_1$ occurred at $r = 4$ oct/s. The stronger DPs were also more sensitive to changes in sweep rate: $2f_1 - f_2$ showed the largest drop in SNR across sweep rates, whereas $3f_1 - 2f_2$ showed the smallest. This indicates that the effect of sweep rate on SNR depends on DP type.

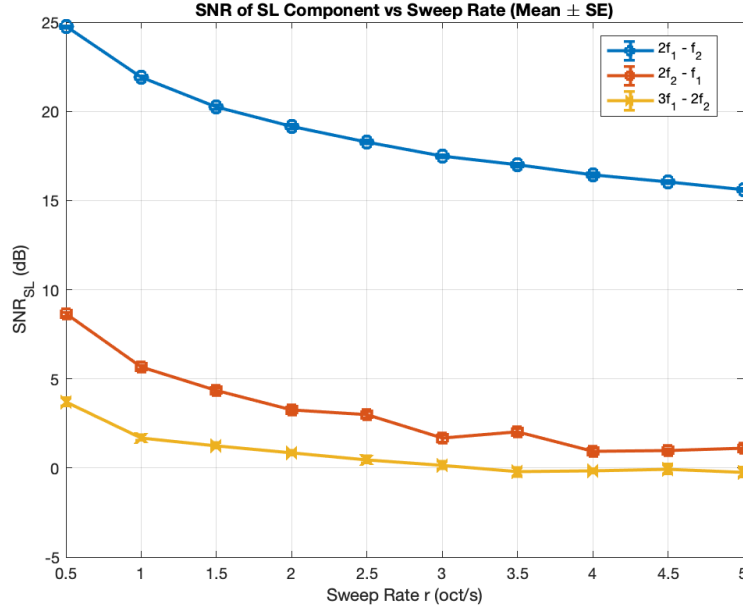


Figure 5.5: Simulation results for Sweep Rate vs. SNR of the SL component(with AWGN).

5.2 Alpha Results

Alpha displayed a pattern similar to that of sweep rate. Increasing α led to increased temporal separation between components, decreased overlap, and lower SNR.

5.2.1 Temporal Separation

As hypothesized, α has a statistically significant effect on temporal separation in both the noise and no-noise simulations, as supported by both the ANOVA results and the post hoc plots. In general, increasing α led to greater temporal separation between the components. The main exception was the $2f_2 - f_1$ DP, which showed little to no meaningful change in component separation as α varied under either condition.

This increase is likely due to the increased separation of the f_1 and f_2 stimulus waves, which pushes the characteristic place of lower-frequency DPs more apically. The inverse behavior was expected for $2f_2 - f_1$, since increasing spacing between stimulus waves pushes higher-frequency DPs more basally, but this was not observed.

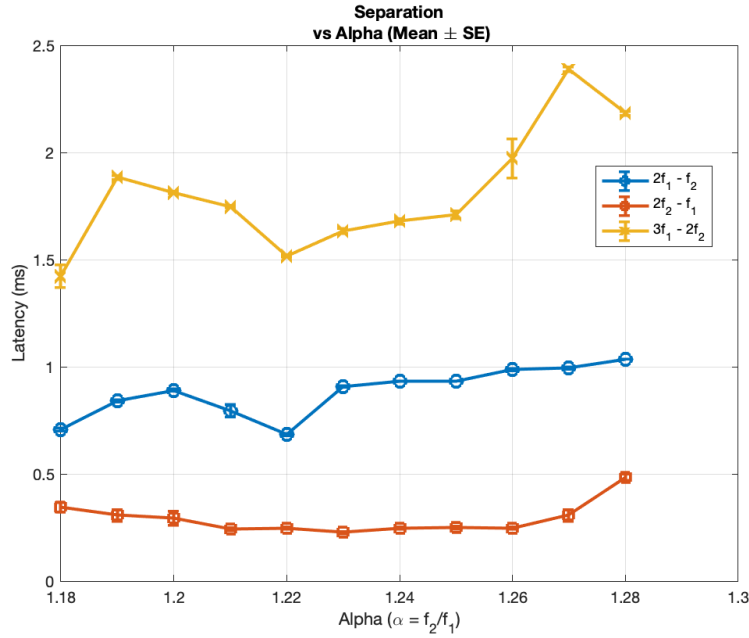


Figure 5.6: Simulation results for alpha vs. Temporal Separation of Components (no AWGN).

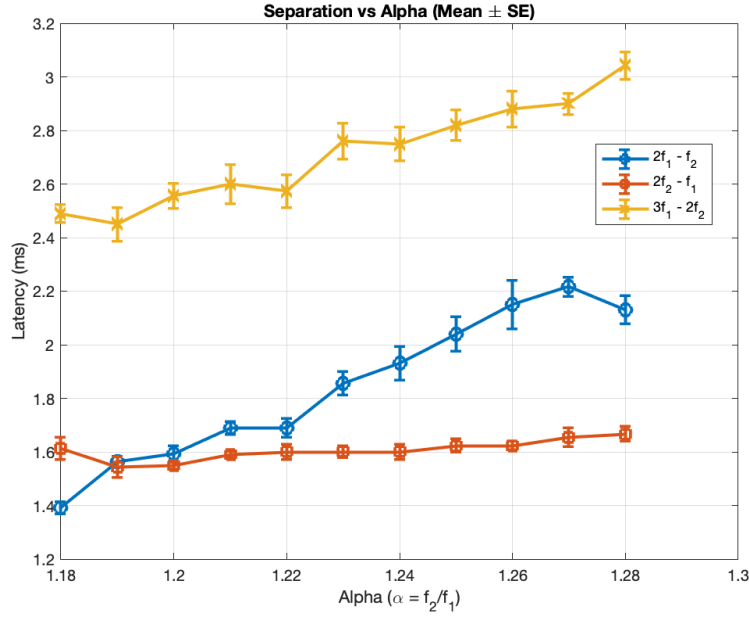


Figure 5.7: Simulation results for alpha vs. Temporal Separation of Components (with AWGN).

5.2.2 Overlap

The F-statistics and p-values confirm that α has a significant effect on temporal overlap and significantly interacts with DP type. Given the effect of α on temporal separation, we would expect the inverse pattern for overlap percentage, which is indeed the case. The overlap percentage of $2f_2 - f_1$ was the least sensitive to changes in α , which reflects the negligible changes in temporal separation for the same DP. The other tested DPs were more responsive and displayed a clearer downward trend in overlap percentage as α increased, reflecting the increase in component separation. This held true in simulations both with and without noise.

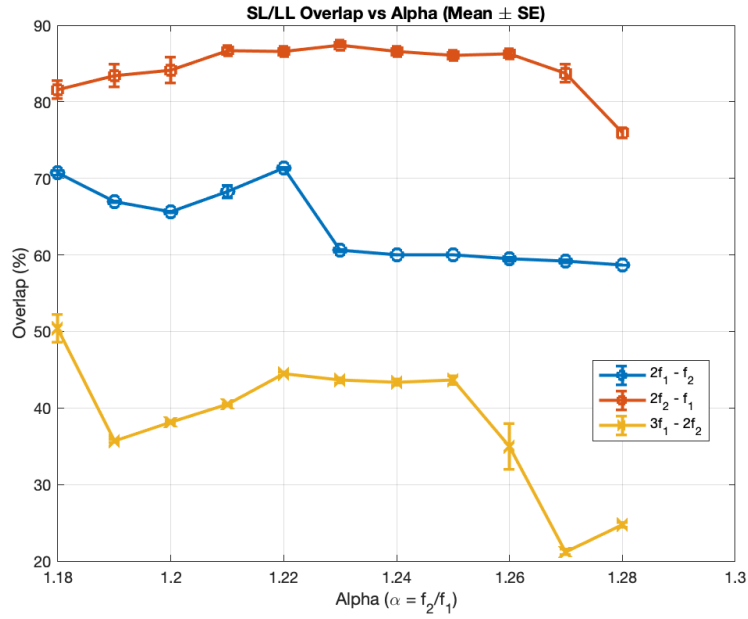


Figure 5.8: Simulation results for alpha vs. Overlap % of Components (no AWGN).

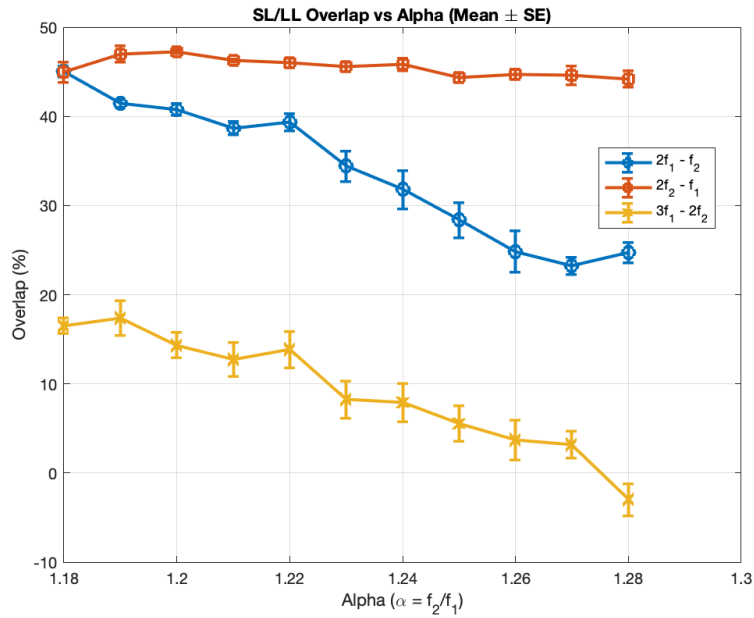


Figure 5.9: Simulation results for alpha vs. Overlap % of Components (with AWGN).

5.2.3 SNR

Based on the plots, there is an obvious pattern: as α increases, the SNR of the SL component decreases. The decline is monotonic for both $2f_1 - f_2$ and $2f_2 - f_1$. The $3f_1 - 2f_2$ shows a steady downward trend, though not a perfectly monotonic one. A likely explanation is that as α increases, the stimulus waves are spread farther apart and interact less. Less interaction between the waves produces a weaker distortion, meaning lower amplitude and therefore lower SNR.

The statistics confirm the significance of these results. They also corroborate a significant relationship between α and DP type. DPs with higher amplitudes exhibit a stronger decline in SNR with increasing α .

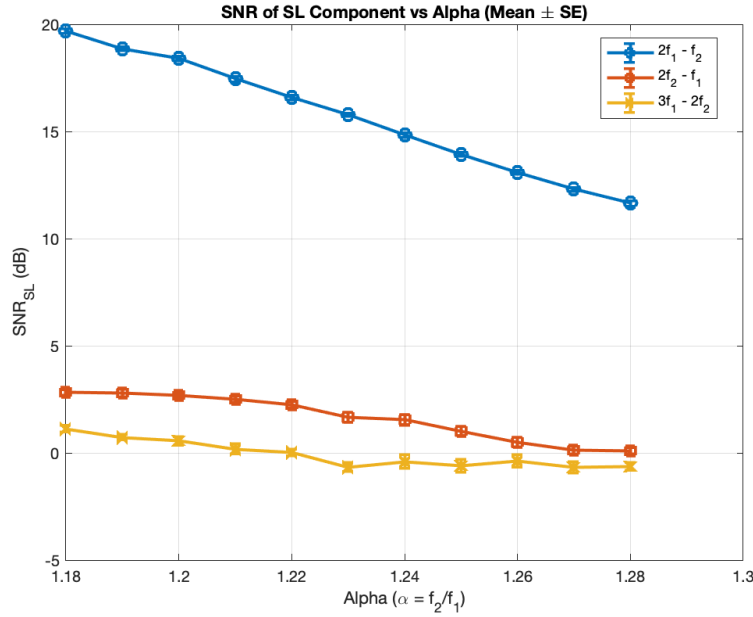


Figure 5.10: Simulation results for alpha vs. SNR of the SL component (with AWGN).

5.3 Stimulus Level Results

Increasing stimulus level resulted in a higher SNR of the short-latency component. However, it also led to decreased temporal separation and increased overlap.

5.3.1 Temporal Separation

Stimulus strength has a clear effect on SL and LL temporal separation: as the stimuli become weaker, separation is enhanced. This effect is statistically significant in both datasets. In simulations without noise, stimulus strength had a strong main effect ($F = 257.5$), and in the noisy simulations it remained significant ($F = 33.96$). Looking at the stimulus-strength means, the shortest latencies occur at the strongest or near-strongest conditions, while weaker conditions show greater separation.

In the clean results, the strongest rise in latency appears for the weaker standard and scissor-paradigm conditions. In the noisy results, the same pattern remains, but all values shift upward, with the strongest condition ($L_1 = 70$ dB, $L_2 = 60$ dB) showing the smallest mean latency and the weakest condition ($L_1 = 55$ dB, $L_2 = 40$ dB) the largest. Thus, weaker stimulus strength makes the SL and LL components more temporally separated, and noise amplifies that separation.

The interaction between stimulus strength and DP type was also highly significant in both cases, so the strength effect is not the same for all DPs. The $2f_1 - f_2$ was the most responsive to changes in stimulus strength. It showed the largest increase in separation with changing stimulus strength in both scenarios. For standard stimulus levels, $2f_2 - f_1$ deviated from the general pattern in both scenarios, showing a decrease in component separation with decreasing stimulus level. The $3f_1 - 2f_2$ did not behave the same across different noise settings. With no noise, it displayed the general trend noted earlier. In noisy settings, the separation exhibited no substantial change across varying stimulus conditions.

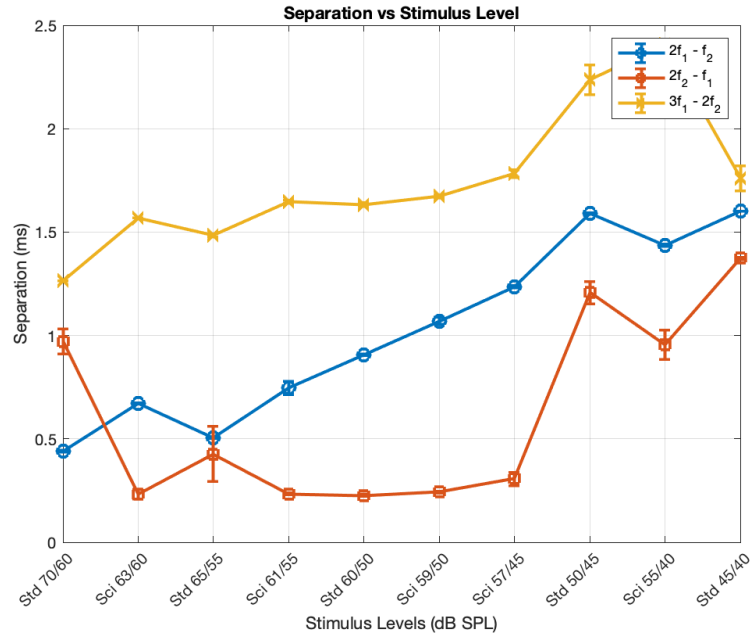


Figure 5.11: Simulation results for Stimulus Strength vs. Temporal Separation of Components (no AWGN).

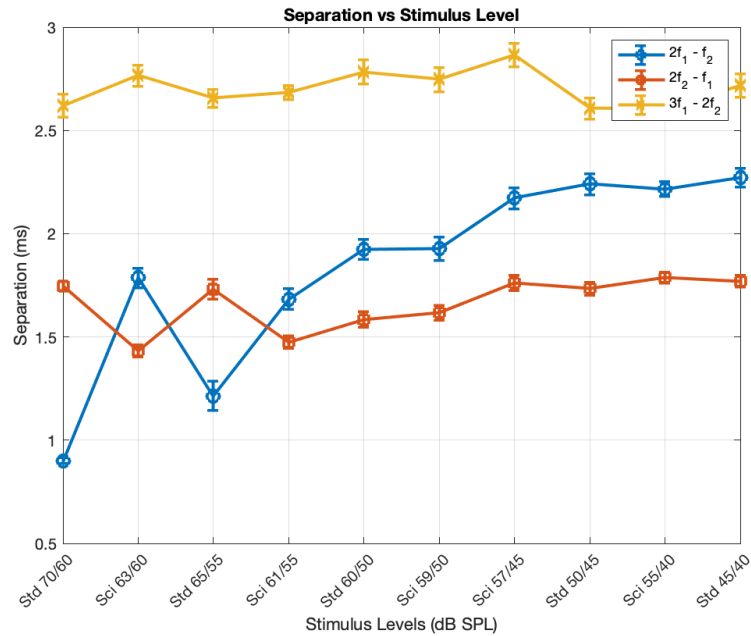


Figure 5.12: Simulation results for Stimulus Strength vs. Temporal Separation of Components (with AWGN).

5.3.2 Overlap

Stimulus strength had a significant effect on overlap, showing that overlap values changed systematically across the different condition levels. In general, stronger stimulus pairings tended to produce higher overlap than weaker pairings, although this pattern was not perfectly uniform across all comparisons. The statistics also show a significant interaction between stimulus level and DP type. Some DP types showed a much stronger increase in overlap as stimulus strength increased, such as $2f_1 - f_2$, whereas others changed more modestly. The $3f_1 - 2f_2$ showed a moderate increase with increasing stimulus levels in noise-free simulations; however, there was no response to changing stimulus intensity in noisy simulations. Similar to temporal separation, $2f_2 - f_1$ deviated from the general trend for standard stimulus levels.

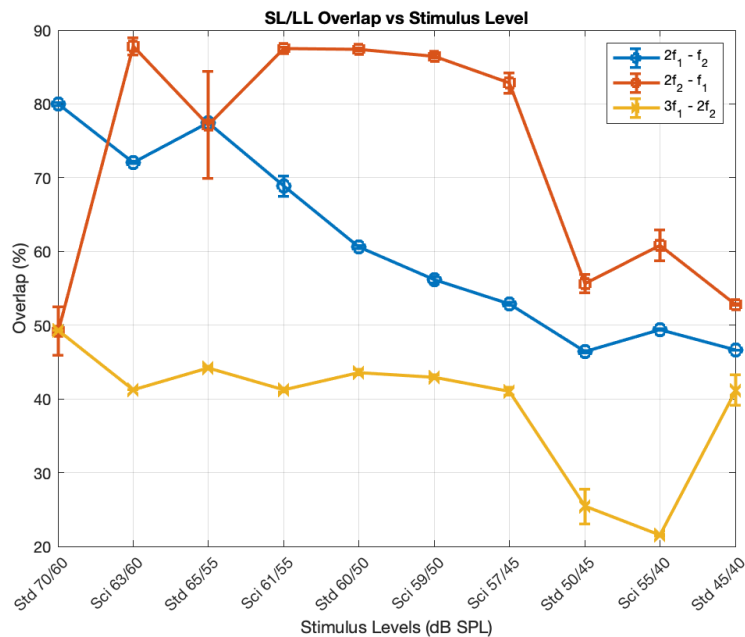


Figure 5.13: Simulation results for Stimulus Results vs. Overlap % of Components (no AWGN).

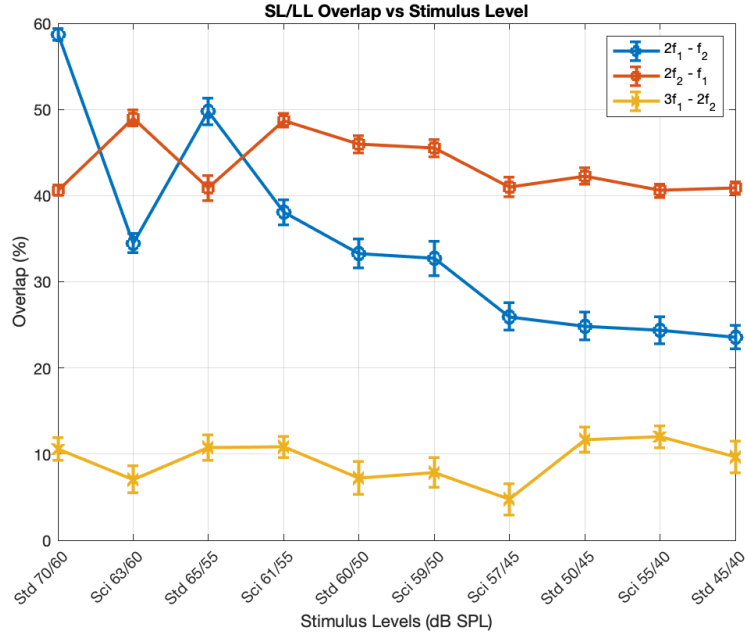


Figure 5.14: Simulation results for Stimulus Strength vs. Overlap % of Components (with AWGN).

5.3.3 SNR

As hypothesized, stimulus intensity had the greatest effect on SNR, which is indicated by having the highest F-statistic of all the metrics. Stronger stimulus conditions produced higher SNR values. The $2f_1 - f_2$ demonstrated the greatest responsiveness to changes in stimulus strength. The strongest stimulus level, $L_1 = 70$ dB and $L_2 = 60$ dB, had the highest SNR, and the weakest stimulus level, $L_1 = 45$ dB and $L_2 = 40$ dB, had the lowest SNR (1.18 dB). It also showed the greatest variance between different stimulus levels. Smaller-amplitude DPs displayed the same tendency to a lesser degree; the weaker the distortion, the less reactive it is to changes.

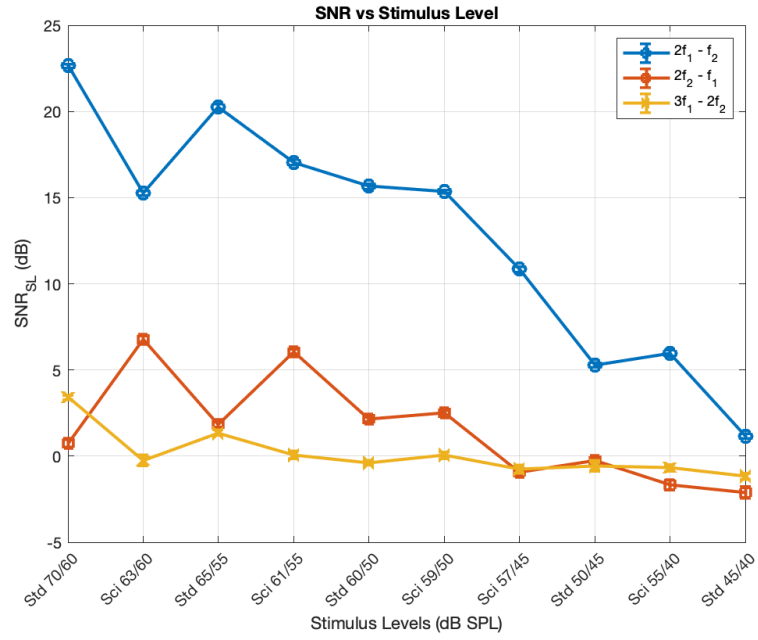


Figure 5.15: Simulation results for Stimulus Strength vs. SNR of the SL component (with AWGN).

Chapter 6

Conclusion and Parameter Recommendations

6.1 Discussion

The objective of this study was to investigate how changes in parameters affected separation of the SL and LL components, signal overlap, and SNR, and to identify optimal parameter values or ranges. All three parameters showed similar trade-offs: improved component separation resulted in lower SNR. This trend was more pronounced in simulations with AWGN and in the cubic distortion product, $2f_1 - f_2$. Because the cubic distortion product is the strongest and most pronounced DPOAE found in humans Vencovský et al. [6], optimal parameters are determined with this specific DP in mind.

Increasing sweep rate led to a subtle increase in temporal separation between the SL and LL components in noisy simulations. In non-noisy simulations, changes in temporal separation were minimal. Based on the statistics, the non-noisy simulations are more significant and may therefore be more reflective of how component separation behaves with changing sweep rate. However, faster sweep rates also cause a reduction in SNR. When creating the recursive windows, the number of windows is scaled proportionally with sweep rate, but this

occurs in combination with decreasing signal duration, which leads to decreased window size. Previous research has suggested that decreasing window size makes measurements significantly more susceptible to noise [10]. Though a sweep rate of 0.5 oct/s did produce the highest SNR, it also had the least separation between the SL and LL components, in addition to being the most time-consuming and computationally demanding. Also, due to inferior component separation, it is possible that the SNR calculations are less accurate. Conversely, a sweep rate of 5 oct/s suffered the greatest amplitude reduction, but did provide improved separation. A middle range from 1.5 oct/s to 3.5 oct/s appears to provide the best trade-off.

Alpha dictates the spacing of the stimulus waves, which can directly affect both component separation and SNR. As previously mentioned in Section 5.2.3, greater spacing between the waves results in less interaction and weaker distortion amplitude. Greater spacing also means that the location of the f_2 wave is shifted basally. Since the distortion is generated closer to the f_2 tonotopic place, the location of the generated distortion is also shifted basally. This, combined with the characteristic frequency of $2f_1 - f_2$ shifting apically, means it takes longer for the reflection component to reach its tonotopic place, ultimately causing greater temporal separation between components. Similar to sweep rate, no single optimal value was identified. However, to obtain the best trade-off between separation and SNR, an alpha range of 1.2 to 1.24 is recommended.

Overall, moderate to high stimulus strengths provided the best trade-off across the outcome measures, which agrees with the hypothesis. Although weaker stimulus levels increased temporal separation between the SL and LL components, this came at the cost of substantially lower SNR, which reduced signal reliability and interpretability. In contrast, the strongest stimulus conditions produced the highest SNR values, but also increased overlap and reduced temporal separation, making component distinction less clear.

Prior research indicates that this trend reflects cochlear behavior. As stimulus intensity increases, the amplitude of the SL component increases, but the contribution of the LL component decreases significantly [7]. This results in less interference between the two sources

and ultimately a higher SNR. However, cochlear latencies generally decrease as stimulus intensity increases. Based on research by Charaziak and Shera [9], the center of gravity of the OAE waveform shifts toward earlier times at higher stimulus levels. This general shortening of delays can result in the components arriving closer together in time and becoming harder to separate.

In combination with previous research, these findings suggest that the most effective operating range lies in the moderate to high stimulus conditions, where SNR remains sufficiently strong while overlap and loss of separation are not excessive. The stimulus levels that best represent this trade-off are:

$L_1 = 63 \text{ dB}, \quad L_2 = 60 \text{ dB}$	(scissor paradigm)
$L_1 = 61 \text{ dB}, \quad L_2 = 55 \text{ dB}$	(scissor paradigm)
$L_1 = 60 \text{ dB}, \quad L_2 = 50 \text{ dB}$	(standard)
$L_1 = 59 \text{ dB}, \quad L_2 = 50 \text{ dB}$	(scissor paradigm)

An L_2 stimulus level greater than 60 dB can be considered too high, and an L_2 stimulus level less than 50 dB can be considered too low.

6.2 Limitations and Future Work

The data were obtained exclusively from a cochlear simulator in MATLAB. While the use of a MATLAB-based cochlear simulator is a well-established method for identifying fundamental cochlear tendencies and testing theoretical methods, the findings must be validated through live testing to ensure accuracy.

Simulations do have practical and research value. Researchers have successfully used active models to reproduce salient features of human otoacoustic emissions, such as DPOAE

fine structure, and to explore complex biomechanical interactions that are difficult to measure directly in a living ear [29]. However, models are inherently simplified and cannot always account for individual physiological variability. For example, the Verhulst cochlear simulator used in this study is a one-dimensional model, even though the cochlea is a three-dimensional structure [40]. With such a constraint, it is difficult to precisely emulate certain cochlear behaviors, such as microscopic source generation or realistic reverse transmission [41]. It is not that these models do not attempt to account for such attributes, but rather that they must rely on approximations.

Also, due to time constraints, only ten simulations were run for each parameter value. More simulations are necessary for each parameter to verify the observed patterns and to ensure that the statistics are robust.

Parameter optimization for the SSS method should also be extended to the hearing impaired. In clinical settings, tests are performed on patients in different age groups and with different conditions. By setting the irregularities off in the `verhulst2012` cochlear model, we can simulate an impaired ear. Using parameters $r = 2.5$ oct/s, $\alpha = 1.22$, $L_1 = 61dB$, and $L_2 = 55dB$, an initial simulation was conducted without AWGN. The SNR outcome exhibited the most variance. A significant decrease was recorded with all three DPs having sub-zero values. $2f_2 - f_1$ reported the lowest SNR, highest overlap, and the lowest temporal separation of components. $3f_1 - 2f_2$ reported the highest SNR, lowest overlap, and greatest temporal separation of components. Although numerical values were not emphasized in the study, it should be noted that test simulation with irregularities off reported similar outcome values to simulations with noise (with the exception of SNR of the $2f_1 - f_2$ DP).

6.3 Final Remarks

Parameter optimization is central to advancing the SSS technique. By identifying optimal ranges for sweep rate, alpha, and stimulus strength, we can improve separation of the SL and LL components while minimizing losses in SNR. These findings contribute to the broader objective of enhancing the SSS technique and help maximize its potential, both for research and for future clinical assessment of cochlear health.

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Appendix A: ANOVA Results

Sweep Rate Simulations (no noise)

ANOVA Results for Temporal Separation

Table 8.1 reports the ANOVA results for Temporal Separation of components for Sweep Rate simulations without AWGN.

Table 8.1: ANOVA Results for Temporal Separation

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Sweep Rate	0.1768	9	0	0.01965	11.26	6.113e-15
DP Type	122.6	2	0	61.31	3.514e+04	0
Sweep Rate:DP Type	0.2089	18	0	0.01161	6.653	9.748e-14
Error	0.4711	270	0	0.001745	—	—
Total	123.5	299	0	—	—	—

ANOVA Results for Overlap

Table 8.2 reports the ANOVA results for overlap percentage of components in sweep rate simulations without AWGN.

Table 8.2: ANOVA Results for Overlap (%)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Sweep Rate	214.7	9	0	23.86	6.382	3.28e-08
DP Type	1.15e+05	2	0	5.748e+04	1.538e+04	7.23e-279
Sweep Rate:DP Type	247	18	0	13.72	3.671	1.512e-06
Error	1009	270	0	3.738	—	—
Total	1.164e+05	299	0	—	—	—

Sweep Rate Simulations with AWGN

ANOVA Results for Temporal Separation

Table 8.3 reports the ANOVA results for temporal separation of components in sweep rate simulations with AWGN.

Table 8.3: ANOVA Results for Temporal Separation

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Sweep Rate	1.31	9	0	0.1455	9.819	5.259e-13
DP Type	68.17	2	0	34.08	2300	2.633e-170
Sweep Rate:DP Type	1.24	18	0	0.06888	4.647	6.405e-09
Error	4.001	270	0	0.01482	—	—
Total	74.72	299	0	—	—	—

ANOVA Results for Overlap (%)

Table 8.4 reports the ANOVA results for the overlap percentage of the components in sweep rate simulations with AWGN.

Table 8.4: ANOVA Results for Overlap (%)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Sweep Rate	930.9	9	0	103.4	8.22	8.392e-11
DP Type	6.378e+04	2	0	3.189e+04	2534	1.075e-175
Sweep Rate:DP Type	1141	18	0	63.38	5.036	7.257e-10
Error	3398	270	0	12.58	—	—
Total	6.925e+04	299	0	—	—	—

ANOVA Results for SNR

Table 8.5 reports the ANOVA results for the SNR of the SL component in sweep rate simulations with AWGN.

Table 8.5: ANOVA Results for SNR SL (dB)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Sweep Rate	1211	9	0	134.5	487.6	2.21e-161
DP Type	1.897e+04	2	0	9484	3.438e+04	0
Sweep Rate:DP Type	162.6	18	0	9.036	32.76	2.263e-57
Error	74.48	270	0	0.2759	—	—
Total	2.042e+04	299	0	—	—	—

Alpha Simulations (no noise)

ANOVA Results for Temporal Separation

Table 8.6 reports the ANOVA results for Temporal Separation of the components in alpha simulations without AWGN.

Table 8.6: ANOVA Results for Temporal Separation

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
alpha	5.745	10	0	0.5745	114.9	7.155e-96
DP Type	129.8	2	0	64.9	1.298e+04	8.668e-290
alpha:DP Type	4.171	20	0	0.2085	41.71	4.842e-74
Error	1.485	297	0	0.005	—	—
Total	141.2	329	0	—	—	—

ANOVA Results for Overlap (%)

Table 8.7 reports the ANOVA results for the overlap percentage in alpha simulations without AWGN.

Table 8.7: ANOVA Results for Overlap (%)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
alpha	6392	10	0	639.2	92.1	6.898e-85
DP Type	1.176e+05	2	0	5.881e+04	8474	1.145e-262
alpha:DP Type	4820	20	0	241	34.73	9.685e-66
Error	2061	297	0	6.94	—	—
Total	1.309e+05	329	0	—	—	—

Alpha Simulations with AWGN

ANOVA Results for Temporal Separation

Table 8.8 reports the ANOVA results for Temporal Separation of components for alpha simulations with AWGN.

Table 8.8: ANOVA Results for Temporal Separation

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
alpha	7.955	10	0	0.7955	37.04	1.268e-46
DP Type	74.61	2	0	37.3	1737	1.253e-164
alpha:DP Type	3.364	20	0	0.1682	7.833	4.896e-18
Error	6.378	297	0	0.02148	—	—
Total	92.3	329	0	—	—	—

ANOVA Results for Overlap (%)

Table 8.9 reports the ANOVA results for overlap percentage of components for alpha simulations with AWGN.

Table 8.9: ANOVA Results for Overlap (%)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
alpha	7264	10	0	726.4	36.8	2.137e-46
DP Type	7.609e+04	2	0	3.805e+04	1928	7.801e-171
alpha:DP Type	2936	20	0	146.8	7.437	4.962e-17
Error	5862	297	0	19.74	—	—
Total	9.215e+04	329	0	—	—	—

ANOVA Results for SNR SL (dB)

Table 8.10 reports the ANOVA results for the SNR of the SL component for alpha simulations with AWGN.

Table 8.10: ANOVA Results for SNR SL (dB)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
alpha	640.8	10	0	64.08	207.5	1.24e-127
DP Type	1.646e+04	2	0	8232	2.666e+04	0
alpha:DP Type	271.7	20	0	13.58	44	1.533e-76
Error	91.7	297	0	0.3088	—	—
Total	1.747e+04	329	0	—	—	—

Stimulus Strength Simulations (no noise)

ANOVA Results for Temporal Separation

Table 8.11 reports the ANOVA results for temporal separation of the components for stimulus strength simulations without AWGN.

Table 8.11: ANOVA Results for Temporal Separation

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Stimulus Strength	33.13	9	0	3.681	257.5	5.613e-127
DP Type	65.36	2	0	32.68	2286	5.795e-170
Stimulus Strength:DP Type	13.01	18	0	0.7226	50.54	1.581e-75
Error	3.86	270	0	0.0143	—	—
Total	115.4	299	0	—	—	—

ANOVA Results for Overlap (%)

Table 8.12 reports the ANOVA results for the overlap percentage for stimulus strength simulations without AWGN.

Table 8.12: ANOVA Results for Overlap (%)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Stimulus Strength	2.483e+04	9	0	2759	96.64	3.74e-79
DP Type	5.816e+04	2	0	2.908e+04	1019	1.62e-126
Stimulus Strength:DP Type	2.018e+04	18	0	1121	39.28	1.086e-64
Error	7707	270	0	28.54	—	—
Total	1.109e+05	299	0	—	—	—

Stimulus Strength Simulations with AWGN

ANOVA Results for Temporal Separation

Table 8.13 reports the ANOVA results for temporal separation of components across stimulus strength levels with AWGN.

Table 8.13: ANOVA Results for Temporal Separation

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Stimulus Strength	7.457	9	0	0.8286	33.96	1.204e-39
DP_Type	63.85	2	0	31.92	1308	1.213e-139
Stimulus Strength:DP_Type	12.47	18	0	0.6926	28.38	6.691e-52
Error	6.588	270	0	0.0244	—	—
Total	90.36	299	0	—	—	—

ANOVA Results for Overlap (%)

Table 8.14 reports the ANOVA results for overlap percentage of components across stimulus strength levels with AWGN.

Table 8.14: ANOVA Results for Overlap (%)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Stimulus Strength	4280	9	0	475.5	20.14	8.618e-26
DP Type	6.508e+04	2	0	3.254e+04	1378	2.079e-142
Stimulus Strength:DP Type	8762	18	0	486.8	20.61	9.411e-41
Error	6376	270	0	23.62	—	—
Total	8.45e+04	299	0	—	—	—

ANOVA Results for SNR SL (dB)

Table 8.15 reports the ANOVA results for the SNR of the SL component across stimulus strength levels with AWGN.

Table 8.15: ANOVA Results for SNR SL (dB)

Source	Sum Sq.	d.f.	Singular?	Mean Sq.	F	Prob>F
Stimulus Strength	2856	9	0	317.3	830	4.139e-191
DP Type	9696	2	0	4848	1.268e+04	1.119e-267
Stimulus Strength:DP Type	2329	18	0	129.4	338.4	1.433e-173
Error	103.2	270	0	0.3823	—	—
Total	1.498e+04	299	0	—	—	—